

Riemanni hüpoteesist

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October 7, 2026

Part I

hüpotees

$$\begin{aligned} & \exists_{x_1} (\\ & \zeta(x_1) = 0 \wedge \\ & x_1 \in C \wedge \\ & \neg(Re(x_1) = 1/2) \wedge \\ & \neg(\exists_{x_2} (x_2 \in N \wedge Re(x_1) = -x_2 * 2 \wedge Im(x_1) = 0) \\ &) \end{aligned}$$

Part II

muud seoud väited

1 zetafunktsioonist

1.1 kui reaalsosa suurem kui 1

$$\forall_{x_1} (x_1 \in C \wedge Re(x_1) < 1 \rightarrow \zeta(-x_1) = \sum_{n=1}^{\infty} (n^{x_1}) = \sum_{n=1}^{\infty} (n^{x_1}) = \sum_{n=1}^{\infty} (n^{Re(x_1)+i*Im(x_1)}) = \sum_{n=1}^{\infty} (n^{Re(x_1)} * n^{i*Im(x_1)}) = \sum_{n=1}^{\infty} (n^{Re(x_1)} * (\cos(Im(x_1) * \ln(n)) + i * \sin(Im(x_1) * \ln(n))))$$

1.2 Dirichlet series

avaldis1: Dirichlet_series_Wikipediast

$$\begin{aligned} & \forall_{x_1} (x_1 \in C \wedge Re(x_1) > 0 \rightarrow \zeta(x_1) = (x_1 - 1)^{-1} * \sum_{n=1}^{\infty} (\frac{n}{(n+1)^{x_1}} - \frac{n^{-x_1}}{n^{x_1}}) \\ & \text{lihtsustan seda avaldist:} \\ & \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\ & \zeta(x_1 + i * x_2) = \\ & (x_1 - 1 + i * x_2)^{-1} * \sum_{n=1}^{\infty} (\frac{n}{(n+1)^{x_1+i*x_2}} - \frac{n^{-x_1-i*x_2}}{n^{x_1+i*x_2}}) = \\ & (x_1 - 1 + i * x_2)^{-1} * \sum_{n=1}^{\infty} (\frac{n}{(n+1)^{x_1} * (\cos(x_2 * \ln(n+1)) + i * \sin(x_2 * \ln(n+1)))} - \frac{n^{-x_1-i*x_2}}{n^{x_1} * (\cos(x_2 * \ln(n)) + i * \sin(x_2 * \ln(n)))}) = \\ & \frac{x_1-1-i*x_2}{(x_1-1)^2+x_2^2} * \sum_{n=1}^{\infty} (n * (n+1)^{-x_1} * (\cos(x_2 * \ln(n+1)) - i * \sin(x_2 * \ln(n+1))) - (n - x_1 - i * x_2) * n^{-x_1} * (\cos(x_2 * \ln(n)) - i * \sin(x_2 * \ln(n)))) = \\ & \sum_{n=1}^{\infty} (((x_1 - 1) - i * x_2) * (\cos(x_2 * \ln(n+1)) - i * \sin(x_2 * \ln(n+1)))) * ((x_1 - 1)^2 + x_2^2)^{-1} * n * (n+1)^{-x_1} - \\ & ((x_1 - 1) - i * x_2) * (\cos(x_2 * \ln(n)) - i * \sin(x_2 * \ln(n))) * ((x_1 - 1)^2 + x_2^2)^{-1} * (n - x_1 - i * x_2) * n^{-x_1}) = \end{aligned}$$

$$\begin{aligned} & \sum_{n=1}^{\infty} (((((x_1 - 1) * \cos(x_2 * \ln(n + 1)) - x_2 * \sin(x_2 * \ln(n + 1))) - i * ((x_1 - 1) * \sin(x_2 * \ln(n + 1)) + x_2 * \cos(x_2 * \ln(n + 1)))) * ((x_1 - 1)^2 + x_2^2)^{-1} * n * (n + 1)^{-x_1} - \\ & ((n - x_1) * ((x_1 - 1) * \cos(x_2 * \ln(n)) - x_2 * \sin(x_2 * \ln(n))) - x_2 * ((x_1 - 1) * \sin(x_2 * \ln(n)) + x_2 * \cos(x_2 * \ln(n))) - i * \\ & ((n - x_1) * ((x_1 - 1) * \sin(x_2 * \ln(n)) + x_2 * \cos(x_2 * \ln(n))) + x_2 * (((x_1 - 1) * \cos(x_2 * \ln(n)) - x_2 * \sin(x_2 * \ln(n)))))) * \\ & ((x_1 - 1)^2 + x_2^2)^{-1} * n^{-x_1}) = \end{aligned}$$

$$\begin{aligned} & \sum_{n=1}^{\infty} (((x_1 - 1) * \cos(x_2 * \ln(n + 1)) - x_2 * \sin(x_2 * \ln(n + 1))) * ((x_1 - 1)^2 + x_2^2)^{-1} * n * (n + 1)^{-x_1} - ((n - x_1) * ((x_1 - 1) * \cos(x_2 * \ln(n)) - x_2 * \sin(x_2 * \ln(n))) - x_2 * ((x_1 - 1) * \sin(x_2 * \ln(n)) + x_2 * \cos(x_2 * \ln(n)))) * ((x_1 - 1)^2 + x_2^2)^{-1} * \\ & n^{-x_1} - i * (((x_1 - 1) * \sin(x_2 * \ln(n + 1)) + x_2 * \cos(x_2 * \ln(n + 1))) * ((x_1 - 1)^2 + x_2^2)^{-1} * n * (n + 1)^{-x_1} + ((n - x_1) * ((x_1 - 1) * \sin(x_2 * \ln(n)) + x_2 * \cos(x_2 * \ln(n))) + x_2 * (((x_1 - 1) * \cos(x_2 * \ln(n)) - x_2 * \sin(x_2 * \ln(n)))) * ((x_1 - 1)^2 + x_2^2)^{-1} * \\ & n^{-x_1})) = \\ & ((x_1 - 1)^2 + x_2^2)^{-1} * \sum_{n=1}^{\infty} (((x_1 - 1) * \cos(x_2 * \ln(n + 1)) - x_2 * \sin(x_2 * \ln(n + 1))) * n * (n + 1)^{-x_1} - ((n - x_1) * ((x_1 - 1) * \cos(x_2 * \ln(n)) - x_2 * \sin(x_2 * \ln(n))) - x_2 * ((x_1 - 1) * \sin(x_2 * \ln(n)) + x_2 * \cos(x_2 * \ln(n)))) * n^{-x_1} - i * ((x_1 - 1)^2 + x_2^2)^{-1} * \sum_{n=1}^{\infty} (((x_1 - 1) * \sin(x_2 * \ln(n + 1)) + x_2 * \cos(x_2 * \ln(n + 1))) * n * (n + 1)^{-x_1} - ((n - x_1) * ((x_1 - 1) * \sin(x_2 * \ln(n)) + x_2 * \cos(x_2 * \ln(n))) + x_2 * (((x_1 - 1) * \cos(x_2 * \ln(n)) - x_2 * \sin(x_2 * \ln(n)))) * n^{-x_1})) = \\ &)) \end{aligned}$$

järgneva väite nimi on zeta2_reaalosa_ja_imaginaarosa_eraldatud:

$$\begin{aligned} & \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\ & \zeta(x_1 + i * x_2) = \\ & ((x_1 - 1)^2 + x_2^2)^{-1} * \sum_{n=1}^{\infty} (\\ & ((x_1 - 1) * \cos(x_2 * \ln(n + 1)) - x_2 * \sin(x_2 * \ln(n + 1))) * n * (n + 1)^{-x_1} - \\ & ((n - x_1) * ((x_1 - 1) * \cos(x_2 * \ln(n)) - x_2 * \sin(x_2 * \ln(n))) - x_2 * ((x_1 - 1) * \sin(x_2 * \ln(n)) + x_2 * \cos(x_2 * \ln(n)))) * n^{-x_1} \\ &) \\ & - i * ((x_1 - 1)^2 + x_2^2)^{-1} * \sum_{n=1}^{\infty} (\\ & ((x_1 - 1) * \sin(x_2 * \ln(n + 1)) + x_2 * \cos(x_2 * \ln(n + 1))) * n * (n + 1)^{-x_1} - \\ & ((n - x_1) * ((x_1 - 1) * \sin(x_2 * \ln(n)) + x_2 * \cos(x_2 * \ln(n))) + x_2 * ((x_1 - 1) * \cos(x_2 * \ln(n)) - x_2 * \sin(x_2 * \ln(n)))) * n^{-x_1} \\ &) = \\ &)) \end{aligned}$$

lihtsustan seda väidet edasi

$$\begin{aligned} & \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\ & \zeta(x_1 + i * x_2) = \end{aligned}$$

$$\begin{aligned} & (x_1^2 - 2 * x_1 + 1 + x_2^2)^{-1} * \sum_{n=1}^{\infty} (\\ & \sin(x_2 * \ln(n) + 0 * (2\pi)/4) * x_2 * (n - 1) * n^{-x_1} - \sin(x_2 * \ln(n + 1) + 0 * (2\pi)/4) * x_2 * n * (n + 1)^{-x_1} \\ & + \sin(x_2 * \ln(n) + 1 * (2\pi)/4) * (x_1^2 - x_1 - n * x_1 + n + x_2^2) * n^{-x_1} \\ & - \sin(x_2 * \ln(n + 1) + 1 * (2\pi)/4) * (1 - x_1) * n * (n + 1)^{-x_1} \\ &) \\ & + i * (x_1^2 - 2 * x_1 + 1 + x_2^2)^{-1} * \sum_{n=1}^{\infty} (\\ & \sin(x_2 * \ln(n) + 1 * (2\pi)/4) * x_2 * (n - 1) * n^{-x_1} - \sin(x_2 * \ln(n + 1) + 1 * (2\pi)/4) * x_2 * n * (n + 1)^{-x_1} \\ & + \sin(x_2 * \ln(n) + 2 * (2\pi)/4) * (x_1^2 - x_1 - n * x_1 + n + x_2^2) * n^{-x_1} \\ & - \sin(x_2 * \ln(n + 1) + 2 * (2\pi)/4) * (1 - x_1) * n * (n + 1)^{-x_1} \\ &) \\ & = \\ & \lim_{M \rightarrow \infty} (\sum_{n=1}^M (\\ & + \sin(x_2 * \ln(n) + 1 * (2\pi)/4) * (x_1^2 - x_1 - n * x_1 + n + x_2^2) * n^{-x_1} \\ & - \sin(x_2 * \ln(n + 1) + 1 * (2\pi)/4) * (1 - x_1) * n * (n + 1)^{-x_1} \\ &) - \sin(x_2 * \ln(M + 1) + 0 * (2\pi)/4) * x_2 * M * (M + 1)^{-x_1} * \\ & (x_1^2 - 2 * x_1 + 1 + x_2^2)^{-1} \\ & + i * \lim_{M \rightarrow \infty} (\sum_{n=1}^M (\\ & \sin(x_2 * \ln(n) + 2 * (2\pi)/4) * (x_1^2 - x_1 - n * x_1 + n + x_2^2) * n^{-x_1} \\ & - \sin(x_2 * \ln(n + 1) + 2 * (2\pi)/4) * (1 - x_1) * n * (n + 1)^{-x_1} \\ &) - \sin(x_2 * \ln(M + 1) + 1 * (2\pi)/4) * x_2 * M * (M + 1)^{-x_1} * \\ & (x_1^2 - 2 * x_1 + 1 + x_2^2)^{-1} = \end{aligned}$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\sum_{n=1}^M (\\
& + \sin(x_2 * \ln(n) + 1 * (2\pi)/4) * (x_1^2 - 2 * x_1 + x_2^2 + 1) * n^{-x_1} \\
& + \sin(x_2 * \ln(n) + 1 * (2\pi)/4) * (1 - x_1) * (n - 1) * n^{-x_1} - \sin(x_2 * \ln(n+1) + 1 * (2\pi)/4) * (1 - x_1) * n * (n+1)^{-x_1} \\
&) - \sin(x_2 * \ln(M+1) + 0 * (2\pi)/4) * x_2 * M * (M+1)^{-x_1} * \\
& (x_1^2 - 2 * x_1 + 1 + x_2^2)^{-1} \\
& + i * \lim_{M \rightarrow \infty} (\sum_{n=1}^M (\\
& + \sin(x_2 * \ln(n) + 2 * (2\pi)/4) * (x_1^2 - 2 * x_1 + x_2^2 + 1) * n^{-x_1} \\
& + \sin(x_2 * \ln(n) + 2 * (2\pi)/4) * (1 - x_1) * (n - 1) * n^{-x_1} - \sin(x_2 * \ln(n+1) + 2 * (2\pi)/4) * (1 - x_1) * n * (n+1)^{-x_1} \\
&) - \sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * x_2 * M * (M+1)^{-x_1} * \\
& (x_1^2 - 2 * x_1 + 1 + x_2^2)^{-1} = \\
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^M (\sin(x_2 * \ln(n) + 1 * (2\pi)/4) * n^{-x_1}) \\
& - (\sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * (1 - x_1) + \sin(x_2 * \ln(M+1) + 0 * (2\pi)/4) * x_2) * M * (M+1)^{-x_1} * ((1 - x_1)^2 + \\
& x_2^2)^{-1}) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^M (\sin(x_2 * \ln(n) + 2 * (2\pi)/4) * n^{-x_1}) \\
& - (\sin(x_2 * \ln(M+1) + 2 * (2\pi)/4) * (1 - x_1) + \sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * x_2) * M * (M+1)^{-x_1} * ((1 - x_1)^2 + \\
& x_2^2)^{-1}) \\
& =
\end{aligned}$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^M (\sin(x_2 * \ln(n) + 1 * (2\pi)/4) * n^{-x_1}) \\
& - (\sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * (1 - x_1) + \sin(x_2 * \ln(M+1) + 0 * (2\pi)/4) * x_2) * M * (M+1)^{-x_1} * (1 + 1/M) * \\
& ((1 - x_1)^2 + x_2^2)^{-1}) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^M (\sin(x_2 * \ln(n) + 2 * (2\pi)/4) * n^{-x_1}) \\
& - (\sin(x_2 * \ln(M+1) + 2 * (2\pi)/4) * (1 - x_1) + \sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * x_2) * M * (M+1)^{-x_1} * (1 + 1/M) * \\
& ((1 - x_1)^2 + x_2^2)^{-1}) \\
& =
\end{aligned}$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^M (\sin(x_2 * \ln(n) + 1 * (2\pi)/4) * n^{-x_1}) \\
& - (\sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * (1 - x_1) + \sin(x_2 * \ln(M+1) + 0 * (2\pi)/4) * x_2) * M^{1-x_1} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^M (\sin(x_2 * \ln(n) + 2 * (2\pi)/4) * n^{-x_1}) \\
& - (\sin(x_2 * \ln(M+1) + 2 * (2\pi)/4) * (1 - x_1) + \sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * x_2) * M^{1-x_1} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) =
\end{aligned}$$

))

Valem zeta9_M_plus_1e_ei_ole:

$$\begin{aligned}
& \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) =
\end{aligned}$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^M (\sin(x_2 * \ln(n) + 1 * (2\pi)/4) * n^{-x_1}) \\
& - (\sin(x_2 * \ln(M) + 1 * (2\pi)/4) * (1 - x_1) + \sin(x_2 * \ln(M) + 0 * (2\pi)/4) * x_2) * M^{1-x_1} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^M (\sin(x_2 * \ln(n) + 2 * (2\pi)/4) * n^{-x_1}) \\
& - (\sin(x_2 * \ln(M) + 2 * (2\pi)/4) * (1 - x_1) + \sin(x_2 * \ln(M) + 1 * (2\pi)/4) * x_2) * M^{1-x_1} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&)
\end{aligned}$$

))

kuivõrd M läheneb lõpmatusele võin igal pool avaldises i asemele panna $\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})$ või $\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi/x_2})$.

$$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \zeta(x_1 + i * x_2) =$$

$$\begin{aligned} & \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (\sin(x_2 * \ln(n) + 1 * (2\pi)/4) * n^{-x_1}) \\ & - (\sin(x_2 * \ln(\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})) + 1 * (2\pi)/4) * (1 - x_1) + \sin(x_2 * \ln(\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})) + \\ & 0 * (2\pi)/4) * x_2) * \text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})^{1-x_1} * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) \\ & + i * \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (\sin(x_2 * \ln(n) + 2 * (2\pi)/4) * n^{-x_1}) \\ & - (\sin(x_2 * \ln(\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})) + 2 * (2\pi)/4) * (1 - x_1) + \sin(x_2 * \ln(\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})) + \\ & 1 * (2\pi)/4) * x_2) * \text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})^{1-x_1} * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) = \end{aligned}$$

$$\begin{aligned} & \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (\sin(x_2 * \ln(n) + 1 * (2\pi)/4) * n^{-x_1}) \\ & - (\sin(x_2 * \ln((e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})) + 1 * (2\pi)/4) * (1 - x_1) + \sin(x_2 * \ln((e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})) + 0 * (2\pi)/4) * \\ & x_2) * (e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})^{1-x_1} * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) \\ & + i * \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (\sin(x_2 * \ln(n) + 2 * (2\pi)/4) * n^{-x_1}) \\ & - (\sin(x_2 * \ln((e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})) + 2 * (2\pi)/4) * (1 - x_1) + \sin(x_2 * \ln((e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})) + 1 * (2\pi)/4) * \\ & x_2) * (e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})^{1-x_1} * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) = \end{aligned}$$

$$\begin{aligned} & \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (\sin(x_2 * \ln(n) + 1 * (2\pi)/4) * n^{-x_1}) \\ & - (\sin((\text{floor}(M) * (x_2/\text{abs}(x_2)) + q_1) * 2\pi + 1 * (2\pi)/4) * (1 - x_1) + \sin((\text{floor}(M) * (x_2/\text{abs}(x_2)) + q_1) * 2\pi + 0 * \\ & (2\pi)/4) * x_2) * (e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi*(1-x_1)}) * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) \\ & + i * \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (\sin(x_2 * \ln(n) + 2 * (2\pi)/4) * n^{-x_1}) \\ & - (\sin((\text{floor}(M) * (x_2/\text{abs}(x_2)) + q_2) * 2\pi + 2 * (2\pi)/4) * (1 - x_1) + \sin((\text{floor}(M) * (x_2/\text{abs}(x_2)) + q_2) * 2\pi + 1 * \\ & (2\pi)/4) * x_2) * (e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi*(1-x_1)}) * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) = \\ &)))) \end{aligned}$$

zeta_funktsioon_2_q_ga:

$$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \zeta(x_1 + i * x_2) =$$

$$\begin{aligned} & \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (\sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1}) \\ & - (\sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + \sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi*(1-x_1)} * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) \\ & + i * \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (\sin(x_2 * \ln(n) + (2\pi) * 2/4) * n^{-x_1}) \\ & - (\sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + \sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi*(1-x_1)} * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) = \end{aligned}$$

))))

Nagu kvantrite abilgi öeldud on kehtib võrrand mistahes q_1 ja q_2 väärtuste puhul, aga kuna pole kindel, et millised väärtused q_1 ja q_2 asemele pannes saab kõige paremal kujul avaldise, siis loon su subsectionites võrrandist erinevaid variante, kus q_1 ja q_2 'el on erinevad väärtused.

1.2.1 integraliga

kasutades seost $a * \cos(t) + b * \sin(t) = \sqrt{a^2 + b^2} * \sin(t + \arctan(\frac{a}{b}))$ lihtsustan eelnevat valemit:

$$\begin{aligned} & \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\ & \zeta(x_1 + i * x_2) = \\ & \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (\sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1}) \\ & - \sin(q_1 * 2\pi + \arctan((1-x_1)/x_2)) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1/2} \\ &) \\ & + i * \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (\sin(x_2 * \ln(n) + (2\pi) * 2/4) * n^{-x_1}) \\ & - \sin(q_2 * 2\pi - \arctan(x_2/(1-x_1))) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1/2} \\ &) \\ &)))) \end{aligned}$$

igalt summa elemendilt lahutan integraali eelmisest elemendist selle elemendine ja liidan maha lahutatud osa peale summerimist uuesti tagasi. Mingil põhjusel, mille chatgpt kirjutas tomib see ainult tingimusel $x_1 < 1$.

$$\begin{aligned} & \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge x_1 < 1 \rightarrow \\ & \zeta(x_1 + i * x_2) = \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (\sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1} - \int_{n-1}^n (dy * \sin(x_2 * \ln(y) + (2\pi) * 1/4) * y^{-x_1})) \\ & + \int_0^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (dy * \sin(x_2 * \ln(y) + (2\pi) * 1/4) * y^{-x_1}) \\ & + \int_{e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi}}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi}-1)} (dy * \sin(x_2 * \ln(y) + (2\pi) * 1/4) * y^{-x_1}) \\ & - \sin(q_1 * 2\pi + \arctan((1-x_1)/x_2)) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1/2} \\ &) \\ & + i * \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (\sin(x_2 * \ln(n) + (2\pi) * 2/4) * n^{-x_1} - \int_{n-1}^n (dy * \sin(x_2 * \ln(y) + (2\pi) * 2/4) * y^{-x_1})) \\ & + \int_0^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (dy * \sin(x_2 * \ln(y) + (2\pi) * 2/4) * y^{-x_1}) \\ & + \int_{e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi}}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi}-1)} (dy * \sin(x_2 * \ln(y) + (2\pi) * 2/4) * y^{-x_1}) \\ & - \sin(q_2 * 2\pi - \arctan(x_2/(1-x_1))) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1/2} \\ &) \\ &)))) \end{aligned}$$

asendan integrali sellega võrdse arhmeetilise avaldisega.

$$\begin{aligned} & \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge x_1 < 1 \rightarrow \\ & \zeta(x_1 + i * x_2) = \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (\\ & \sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1} \\ & - (n^{1-x_1} * \sin(x_2 * \ln(n) + \arctan((1-x_1)/x_2)) - (n-1)^{1-x_1} * \sin(x_2 * \ln(n-1) + \arctan((1-x_1)/x_2))) * ((1-x_1)^2 + x_2^2)^{-1/2} \\ &) \\ & + ((e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})^{1-x_1} * \sin(x_2 * \ln(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi}) + \arctan((1-x_1)/x_2)) - 0^{1-x_1} * \sin(x_2 * \ln(0) + \arctan((1-x_1)/x_2))) * ((1-x_1)^2 + x_2^2)^{-1/2} \\ & + ((\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi}-1))^{1-x_1} * \sin(x_2 * \ln(\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi}-1)) + \arctan((1-x_1)/x_2)) - (e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})^{1-x_1} * \sin(x_2 * \ln(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi}) + \arctan((1-x_1)/x_2))) * ((1-x_1)^2 + x_2^2)^{-1/2} \\ &) \\ &)))) \end{aligned}$$

$$\begin{aligned}
& - \sin(q_1 * 2\pi + \arctan((1-x_1)/x_2)) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (\\
& \sin(x_2 * \ln(n) + (2\pi) * 2/4) * n^{-x_1} - \\
& (n^{1-x_1} * \sin(x_2 * \ln(n) - \arctan(x_2/(1-x_1))) + (n-1)^{1-x_1} * \sin(x_2 * \ln(n-1) - \arctan(x_2/(1-x_1)))) * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&) \\
& + ((e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})^{1-x_1} * \sin(x_2 * \ln(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi}) - \arctan(x_2/(1-x_1))) + 0^{1-x_1} * \sin(x_2 * \ln(0) - \arctan(x_2/(1-x_1)))) * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + ((\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi}) - 1)^{1-x_1} * \sin(x_2 * \ln(\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi}) - 1) - \arctan(x_2/(1-x_1)))) + (e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})^{1-x_1} * \sin(x_2 * \ln(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi}) - \arctan(x_2/(1-x_1)))) * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& - \sin(q_2 * 2\pi - \arctan(x_2/(1-x_1))) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&) \\
&))))
\end{aligned}$$

lihtsustan. Kuna kui $x_1 < 1$ siis $\lim_{y \rightarrow 0^+} (y^{1-x_1} * \sin(x_2 * \ln(y) + \arctan((1-x_1)/x_2))) = 0$ ($0^{1-x_1} * \sin(x_2 * \ln(0) + \arctan((1-x_1)/x_2)) = 0$), sest $\sin(x_2 * \ln(0) + \arctan((1-x_1)/x_2))$ on määratu, aga kindlalt piirides -1 kuni 1.

Kuna kui $x_1 < 1$ siis $\lim_{y \rightarrow 0^+} (y^{1-x_1} * \sin(x_2 * \ln(y) + \arctan(x_2/(1-x_1)))) = 0$ ($0^{1-x_1} * \sin(x_2 * \ln(0) - \arctan(x_2/(1-x_1))) = 0$), sest $\sin(x_2 * \ln(0) - \arctan(x_2/(1-x_1)))$ on määratu, aga kindlalt piirides -1 kuni 1.

SIIN OLI VIGA! kas nüüd korras?

$$\begin{aligned}
& \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge x_1 < 1 \rightarrow \\
& \zeta(x_1 + i * x_2) = \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (\\
& \sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1} \\
& - \sin(x_2 * \ln(n) + \arctan((1-x_1)/x_2)) * n^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + \sin(x_2 * \ln(n-1) + \arctan((1-x_1)/x_2)) * (n-1)^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&) \\
& + \sin(q_1 * 2\pi + \arctan((1-x_1)/x_2)) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + ((\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi}) - 1)^{1-x_1} * \sin(x_2 * \ln(\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi}) - 1) + \arctan((1-x_1)/x_2)) - \\
& (e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})^{1-x_1} * \sin(x_2 * \ln(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi}) + \arctan((1-x_1)/x_2))) * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& - \sin(q_1 * 2\pi + \arctan((1-x_1)/x_2)) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (\\
& \sin(x_2 * \ln(n) + (2\pi) * 2/4) * n^{-x_1} \\
& - \sin(x_2 * \ln(n) - \arctan(x_2/(1-x_1))) * n^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + \sin(x_2 * \ln(n-1) - \arctan(x_2/(1-x_1))) * (n-1)^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&) \\
& + \sin(q_2 * 2\pi - \arctan(x_2/(1-x_1))) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + ((\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi}) - 1)^{1-x_1} * \sin(x_2 * \ln(\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi}) - 1) - \arctan(x_2/(1-x_1)))) + (e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})^{1-x_1} * \sin(q_2 * 2\pi - \arctan(x_2/(1-x_1)))) * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& - \sin(q_2 * 2\pi - \arctan(x_2/(1-x_1))) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&) \\
&))))
\end{aligned}$$

taandan

$$\begin{aligned}
& \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge x_1 < 1 \rightarrow \\
& \zeta(x_1 + i * x_2) = \lim_{M \rightarrow \infty} (
\end{aligned}$$

$$\begin{aligned}
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (\\
& \sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1} \\
& - \sin(x_2 * \ln(n) + \arctan((1-x_1)/x_2)) * n^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + \sin(x_2 * \ln(n-1) + \arctan((1-x_1)/x_2)) * (n-1)^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&) \\
& + (\\
& \sin(x_2 * \ln(\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi} - 1)) + \arctan((1-x_1)/x_2)) * (\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi} - 1))^{1-x_1} \\
& - \sin(x_2 * \ln(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi}) + \arctan((1-x_1)/x_2)) * (e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})^{1-x_1} \\
&) * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (\\
& \sin(x_2 * \ln(n) + (2\pi) * 2/4) * n^{-x_1} \\
& - \sin(x_2 * \ln(n) - \arctan(x_2/(1-x_1))) * n^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + \sin(x_2 * \ln(n-1) - \arctan(x_2/(1-x_1))) * (n-1)^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&) \\
& + (\\
& \sin(x_2 * \ln(\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi} - 1) - \arctan(x_2/(1-x_1))) * (\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi} - 1))^{1-x_1} \\
& + \sin(q_2 * 2\pi - \arctan(x_2/(1-x_1))) * (e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})^{1-x_1} \\
&) * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&) \\
&))))
\end{aligned}$$

kuna kui M on väga suur, siis $\lim_{M \rightarrow 0} (\ln(M-2) - \ln(M)) = 0 \wedge \lim_{M \rightarrow 0} (M^x - (M-2)^x) = 0$, siis

$$\begin{aligned}
& \sin(x_2 * \ln(\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi} - 1)) + \arctan((1-x_1)/x_2)) * (\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi} - 1))^{1-x_1} \\
& - \sin(x_2 * \ln(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi}) + \arctan((1-x_1)/x_2)) * (e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})^{1-x_1} = 0
\end{aligned}$$

ja

$$\begin{aligned}
& \sin(x_2 * \ln(\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi} - 1) - \arctan(x_2/(1-x_1))) * (\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi} - 1))^{1-x_1} \\
& + \sin(q_2 * 2\pi - \arctan(x_2/(1-x_1))) * (e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})^{1-x_1} = 0
\end{aligned}$$

seega:

$$\begin{aligned}
& \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge x_1 < 1 \rightarrow \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} (\sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (\\
& \sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1} \\
& - \sin(x_2 * \ln(n) + \arctan((1-x_1)/x_2)) * n^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + \sin(x_2 * \ln(n-1) + \arctan((1-x_1)/x_2)) * (n-1)^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&)) \\
& + i * \lim_{M \rightarrow \infty} (\sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (\\
& \sin(x_2 * \ln(n) + (2\pi) * 2/4) * n^{-x_1} \\
& - \sin(x_2 * \ln(n) - \arctan(x_2/(1-x_1))) * n^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + \sin(x_2 * \ln(n-1) - \arctan(x_2/(1-x_1))) * (n-1)^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&)) \\
&))))
\end{aligned}$$

kuna vastavate q_1 ja q_2 väärtuste valimise korral saab summa piiri suvaliselt nihutada, siis võib summa piiriks lihtsalt M_i panna

$$\begin{aligned}
& \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge x_1 < 1 \rightarrow \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} (\sum_{n=1}^M (\\
& \sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1} \\
& - \sin(x_2 * \ln(n) + \arctan((1-x_1)/x_2)) * n^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + \sin(x_2 * \ln(n-1) + \arctan((1-x_1)/x_2)) * (n-1)^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&)) \\
& + i * \lim_{M \rightarrow \infty} (\sum_{n=1}^M (\\
& - \sin(x_2 * \ln(n)) * n^{-x_1}
\end{aligned}$$

$$\begin{aligned}
& -\sin(x_2 * \ln(n) - \arctan(x_2/(1-x_1))) * n^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + \sin(x_2 * \ln(n-1) - \arctan(x_2/(1-x_1))) * (n-1)^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&)) \\
&))
\end{aligned}$$

Muudan summa piiri 1est 0iks:

$$\begin{aligned}
& \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge x_1 < 1 \rightarrow \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} (\sum_{n=0}^M (\\
& \sin(x_2 * \ln(n+1) + (2\pi) * 1/4) * (n+1)^{-x_1} \\
& - \sin(x_2 * \ln(n+1) + \arctan((1-x_1)/x_2)) * (n+1)^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + \sin(x_2 * \ln(n) + \arctan((1-x_1)/x_2)) * n^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&)) \\
& + i * \lim_{M \rightarrow \infty} (\sum_{n=0}^M (\\
& - \sin(x_2 * \ln(n+1)) * (n+1)^{-x_1} \\
& - \sin(x_2 * \ln(n+1) - \arctan(x_2/(1-x_1))) * (n+1)^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + \sin(x_2 * \ln(n) - \arctan(x_2/(1-x_1))) * n^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&)) \\
&))
\end{aligned}$$

lihtsustan

$$\begin{aligned}
& \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge x_1 < 1 \rightarrow \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} (\sum_{n=0}^M (\\
& \sin(x_2 * \ln(n+1) + (2\pi) * 1/4) * (n+1)^{-x_1} \\
& - \sin(x_2 * \ln(n+1) + \arctan((1-x_1)/x_2)) * (n+1)^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + \sin(x_2 * \ln(n) + \arctan((1-x_1)/x_2)) * n^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&)) \\
& - i * \lim_{M \rightarrow \infty} (\sum_{n=0}^M (\\
& \sin(x_2 * \ln(n+1)) * (n+1)^{-x_1} \\
& - \sin(x_2 * \ln(n+1) + \arctan((1-x_1)/x_2)) * (n+1)^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
& + \sin(x_2 * \ln(n) + \arctan((1-x_1)/x_2)) * n^{1-x_1} * ((1-x_1)^2 + x_2^2)^{-1/2} \\
&)) \\
&))
\end{aligned}$$

1.2.2 edasine lihtsustamine lihtsamaid funktsioone sisaldama.

asendan siinuse selle Tayloriga reaga:

$$\begin{aligned}
& \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} (\\
& \sum_{n_0=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (\\
& n_0^{-x_1} * \sum_{n_1=0}^{\infty} ((-1)^{n_1} * (x_2 * \ln(n_0))^{2*n_1} * \text{factorial}(2 * n_1)^{-1}) \\
&) \\
& - (\sin((q_1 + 1/4) * 2\pi) * (1-x_1) + \sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& - \sum_{n_0=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (\\
& n_0^{-x_1} * \sum_{n_1=0}^{\infty} ((-1)^{n_1} * (x_2 * \ln(n_0))^{2*n_1+1} * \text{factorial}(2 * n_1 + 1)^{-1}) \\
&) \\
& - (\sin((q_2 + 2/4) * 2\pi) * (1-x_1) + \sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) \\
&))))
\end{aligned}$$

muudan tegurite järjekorda:

$$\begin{aligned}
& \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} (
\end{aligned}$$

$$\begin{aligned}
& \sum_{n_0=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (n_0^{-x_1} * \\
& \sum_{n_1=0}^{\infty} ((-1)^{n_1} * x_2^{2*n_1} * \text{factorial}(2*n_1)^{-1} * \ln(n_0)^{2*n_1}) \\
&) \\
& - (\sin((q_1+1/4)*2\pi) * (1-x_1) + \sin((q_1+0/4)*2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& - \sum_{n_0=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (n_0^{-x_1} * \\
& \sum_{n_1=0}^{\infty} ((-1)^{n_1} * x_2^{2*n_1+1} * \text{factorial}(2*n_1+1)^{-1} * \ln(n_0)^{2*n_1+1}) \\
&) \\
& - (\sin((q_2+2/4)*2\pi) * (1-x_1) + \sin((q_2+1/4)*2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) \\
&))))
\end{aligned}$$

asendal naturallogaritmi summaga:

$$\begin{aligned}
& \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} (\\
& \sum_{n_0=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (n_0^{-x_1} * \\
& \sum_{n_1=0}^{\infty} ((-1)^{n_1} * x_2^{2*n_1} * \text{factorial}(2*n_1)^{-1} * (\sum_{n_2=0}^{\infty} ((2*n_2+1)^{-1} * (n_0-1)^{2*n_2+1} * (n_0+1)^{-2*n_2-1}))^{2*n_1} * 2^{2*n_1}) \\
&) \\
& - (\sin((q_1+1/4)*2\pi) * (1-x_1) + \sin((q_1+0/4)*2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n_0=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (n_0^{-x_1} * \sum_{n_1=0}^{\infty} ((\sum_{n_2=0}^{\infty} ((2*n_2+1)^{-1} * (n_0-1)^{2*n_2+1} * (n_0+1)^{-2*n_2-1}))^{2*n_1+1} * x_2^{2*n_1+1} * \\
& \text{factorial}(2*n_1+1)^{-1} * 2^{2*n_1+1} * (-1)^{n_1}) \\
&) \\
& - (\sin((q_2+2/4)*2\pi) * (1-x_1) + \sin((q_2+1/4)*2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) \\
&))))
\end{aligned}$$

kasutan multinomial valemit:

$$\begin{aligned}
& \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} (\\
& \sum_{n_0=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi})} (n_0^{-x_1} * \sum_{n_1=0}^{M-1} (x_2^{2*n_1} * 2^{2*n_1} * (-1)^{n_1} * \sum_{n_2=0}^{(2*n_1+1)^M-1} (\\
& (\sum_{n_3=0}^{M-1} (\text{floor}(n_2/(2*n_1+1)^{n_3}) \% (2*n_1+1)) = 2*n_1) * \\
& \Pi_{n_3=0}^{M-1} (\text{factorial}(\text{floor}(n_2/(2*n_1+1)^{n_3}) \% (2*n_1+1))^{-1} * ((2*n_3+1)^{-1} * (n_0-1)^{2*n_3+1} * (n_0+1)^{-2*n_3-1})^{\text{floor}(n_2/(2*n_1+1)^{n_3}) \% (2*n_1+1)} \\
&)))) \\
& - (\sin((q_1+1/4)*2\pi) * (1-x_1) + \sin((q_1+0/4)*2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_1/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n_0=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})} (n_0^{-x_1} * \sum_{n_1=0}^{M-1} (x_2^{2*n_1+1} * 2^{2*n_1+1} * (-1)^{n_1} * \sum_{n_2=0}^{(2*n_1+2)^M-1} (\\
& (\sum_{n_3=0}^{M-1} (\text{floor}(n_2/(2*n_1+2)^{n_3}) \% (2*n_1+2)) = 2*n_1+1) * \\
& \Pi_{n_3=0}^{M-1} (\text{factorial}(\text{floor}(n_2/(2*n_1+2)^{n_3}) \% (2*n_1+2))^{-1} * ((2*n_3+1)^{-1} * (n_0-1)^{2*n_3+1} * (n_0+1)^{-2*n_3-1})^{\text{floor}(n_2/(2*n_1+2)^{n_3}) \% (2*n_1+2)} \\
&)))) \\
& - (\sin((q_2+2/4)*2\pi) * (1-x_1) + \sin((q_2+1/4)*2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) \\
&))))
\end{aligned}$$

lihtsustan teades, et korutamise asemel võib astmd liita.

$$\begin{aligned}
& \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} (
\end{aligned}$$

$$\begin{aligned}
& \sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})} (n_0^{-x_1} * \sum_{n_1=0}^{M-1} (x_2^{2*n_1} * 2^{2*n_1} * (-1)^{n_1} * \sum_{n_2=0}^{(2*n_1+1)^{M-1}} (\\
& (\sum_{n_3=0}^{M-1} (floor(n_2/(2*n_1+1)^{n_3}) \% (2*n_1+1)) = 2*n_1) * \\
& (n_0-1))^{\sum_{n_3=0}^{M-1} ((2*n_3+1)*(floor(n_2/(2*n_1+1)^{n_3}) \% (2*n_1+1)))} * \\
& (n_0+1))^{-\sum_{n_3=0}^{M-1} ((2*n_3+1)*(floor(n_2/(2*n_1+1)^{n_3}) \% (2*n_1+1)))} * \\
& \Pi_{n_3=0}^{M-1} (factorial(floor(n_2/(2*n_1+1)^{n_3}) \% (2*n_1+1))^{-1} * (2*n_3+1)^{-floor(n_2/(2*n_1+1)^{n_3}) \% (2*n_1+1)} \\
&)) \\
&))) \\
& -(\sin((q_1+1/4)*2\pi)*(1-x_1) + \sin((q_1+0/4)*2\pi)*x_2) * e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})} (n_0^{-x_1} * \sum_{n_1=0}^{M-1} (x_2^{2*n_1+1} * 2^{2*n_1+1} * (-1)^{n_1} * \sum_{n_2=0}^{(2*n_1+2)^{M-1}} (\\
& (\sum_{n_3=0}^{M-1} (floor(n_2/(2*n_1+2)^{n_3}) \% (2*n_1+2)) = 2*n_1+1) * \\
& (n_0-1))^{\sum_{n_3=0}^{M-1} ((2*n_3+1)*(floor(n_2/(2*n_1+2)^{n_3}) \% (2*n_1+2)))} * \\
& (n_0+1))^{-\sum_{n_3=0}^{M-1} ((2*n_3+1)*(floor(n_2/(2*n_1+2)^{n_3}) \% (2*n_1+2)))} * \\
& \Pi_{n_3=0}^{M-1} (factorial(floor(n_2/(2*n_1+2)^{n_3}) \% (2*n_1+2))^{-1} * (2*n_3+1)^{-floor(n_2/(2*n_1+2)^{n_3}) \% (2*n_1+2)} \\
&)) \\
&))) \\
& -(\sin((q_2+2/4)*2\pi)*(1-x_1) + \sin((q_2+1/4)*2\pi)*x_2) * e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) \\
&))))
\end{aligned}$$

selle asemel et üle erinevate n2'ede ristsummade summeeridaarvutan lihtsalt n2'e ristsumma:

SIIN VIGA!

$$\begin{aligned}
& \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} (\\
& \sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})} (n_0^{-x_1} * \sum_{n_1=0}^{(2*M)^{M-1}} (\\
& (\sum_{n_2=0}^{M-1} (floor(n_1/(2*M)^{n_2}) \% (2*M)) \% 2) * \\
& (\sum_{k_2=0}^{M-1} (floor(k_1/(2*M-1)^{k_2}) \% (2*M-1)) < 2*M-1) * \\
& (y * 2)^{\sum_{n_2=0}^{M-1} (floor(n_1/(2*M)^{n_2}) \% (2*M))} * \\
& (-1)^{\sum_{n_2=0}^{M-1} (floor(n_1/(2*M)^{n_2}) \% (2*M)) / 2} * \\
& (x-1)^{\sum_{n_2=0}^{M-1} ((2*n_2+1)*(floor(n_1/(2*M)^{n_2}) \% (2*M)))} * \\
& (x+1)^{-\sum_{n_2=0}^{M-1} ((2*n_2+1)*(floor(n_1/(2*M)^{n_2}) \% (2*M)))} * \\
& \Pi_{n_2=0}^{M-1} (factorial(floor(n_1/(2*M)^{n_2}) \% (2*M))^{-1} * (2*n_2+1)^{-floor(n_1/(2*M)^{n_2}) \% (2*M)} \\
&)) \\
& -(\sin((q_1+1/4)*2\pi)*(1-x_1) + \sin((q_1+0/4)*2\pi)*x_2) * e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})} (n_0^{-x_1} * \sum_{n_1=0}^{(2*M)^{M-1}} (\\
& (\sum_{n_2=0}^{M-1} (floor(n_1/(2*M)^{n_2}) \% (2*M)) \% 2) * \\
& (\sum_{n_2=0}^{M-1} (floor(n_1/(2*M)^{n_2}) \% (2*M)) < 2*M) * \\
& (x_2 * 2)^{\sum_{n_2=0}^{M-1} (floor(n_1/(2*M)^{n_2}) \% (2*M))} * \\
& (-1)^{(\sum_{n_2=0}^{M-1} (floor(n_1/(2*M)^{n_2}) \% (2*M)) + 3) / 2} * \\
& (n_0-1)^{\sum_{n_2=0}^{M-1} ((2*n_2+1)*(floor(n_1/(2*M)^{n_2}) \% (2*M)))} * \\
& (n_0+1)^{-\sum_{n_2=0}^{M-1} ((2*n_2+1)*(floor(n_1/(2*M)^{n_2}) \% (2*M)))} * \\
& \Pi_{n_2=0}^{M-1} (factorial(floor(n_1/(2*M)^{n_2}) \% (2*M))^{-1} * (2*n_2+1)^{-floor(n_1/(2*M)^{n_2}) \% (2*M)} \\
&)) \\
&)
\end{aligned}$$

$$- (\sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + \sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2) + q_2/x_2) * 2\pi * (1 - x_1)} * ((1 - x_1)^2 + x_2^2)^{-1} \\) \\))))$$

kuna Mi lähenedes lõpmatusse see avaldis nagu nii koondub ja selle muutuse tegemine ainult lisab liikmeid, mis on absoluutväärtuselt väiksemad kui oleksid liikmed, mis lisanduksid Mi suurendamisel:

$$\forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\ \zeta(x_1 + i * x_2) = \\ \lim_{M \rightarrow \infty} (\\ \sum_{n_0=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2) + q_1/x_2) * 2\pi})} (n_0^{-x_1} * \sum_{n_1=0}^{(2*M)^{M-1}} (\\ (\sum_{n_2=0}^{M-1} ((\text{floor}(n_1/(2*M-1)^{n_2}) \% (2*M-1)) + 1) \% 2) * \\ (y * 2)^{\sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2*M-1)^{n_2}) \% (2*M-1))} * \\ (-1)^{\sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2*M-1)^{n_2}) \% (2*M-1))/2} * \\ (n_0 - 1)^{\sum_{n_2=0}^{M-1} ((2*n_2+1) * (\text{floor}(n_1/(2*M-1)^{n_2}) \% (2*M-1)))} * \\ (n_0 + 1)^{-\sum_{n_2=0}^{M-1} ((2*n_2+1) * (\text{floor}(n_1/(2*M-1)^{n_2}) \% (2*M-1)))} * \\ \Pi_{n_2=0}^{M-1} (\text{factorial}(\text{floor}(n_1/(2*M-1)^{n_2}) \% (2*M-1))^{-1} * (2*n_2+1)^{-\text{floor}(n_1/(2*M-1)^{n_2}) \% (2*M-1))} \\))/2 \\ - (\sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + \sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2) + q_1/x_2) * 2\pi * (1 - x_1)} * ((1 - x_1)^2 + x_2^2)^{-1} \\) \\ + i * \lim_{M \rightarrow \infty} (\\ \sum_{n_0=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2) + q_2/x_2) * 2\pi})} (n_0^{-x_1} * \sum_{n_1=0}^{(2*M)^{M-1}} (\\ (\sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)) \% 2) * \\ (x_2 * 2)^{\sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M))} * \\ (-1)^{(\sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)) + 3)/2} * \\ (n_0 - 1)^{\sum_{n_2=0}^{M-1} ((2*n_2+1) * (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)))} * \\ (n_0 + 1)^{-\sum_{n_2=0}^{M-1} ((2*n_2+1) * (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)))} * \\ \Pi_{n_2=0}^{M-1} (\text{factorial}(\text{floor}(n_1/(2*M)^{n_2}) \% (2*M))^{-1} * (2*n_2+1)^{-\text{floor}(n_1/(2*M)^{n_2}) \% (2*M))} \\)) \\ - (\sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + \sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2) + q_2/x_2) * 2\pi * (1 - x_1)} * ((1 - x_1)^2 + x_2^2)^{-1} \\) \\))))$$

viin $n_0^{-x_1}$ summa sisse ja panen n_0 summa 0 ist algama.

$$\forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\ \zeta(x_1 + i * x_2) = \\ \lim_{M \rightarrow \infty} (\\ \sum_{n_0=0}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2) + q_1/x_2) * 2\pi}) - 1} (\sum_{n_1=0}^{(2*M)^{M-1}} (\\ ((1 + \%2) * \\ (x_2 * 2)^{\sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M))} * \\ (-1)^{\sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M))/2} * \\ \sum_{n_2=0}^{M-1} ((2*n_2+1) * (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)))} * \\ n_0 \\ (n_0 + 1)^{-x_1} * \\ (n_0 + 2)^{-\sum_{n_2=0}^{M-1} ((2*n_2+1) * (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)))} * \\ \Pi_{n_2=0}^{M-1} (\text{factorial}(\text{floor}(n_1/(2*M)^{n_2}) \% (2*M))^{-1} * (2*n_2+1)^{-\text{floor}(n_1/(2*M)^{n_2}) \% (2*M))} \\)) \\ - (\sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + \sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(\text{floor}(M)/\text{abs}(x_2) + q_1/x_2) * 2\pi * (1 - x_1)} * ((1 - x_1)^2 + x_2^2)^{-1} \\) \\ + i * \lim_{M \rightarrow \infty} (\\ \sum_{n_0=0}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2) + q_2/x_2) * 2\pi}) - 1} (\sum_{n_1=0}^{(2*M)^{M-1}} ($$

$$\begin{aligned}
& (\sum_{n_2=0}^{M-1} (\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M)) \% 2) * \\
& (x_2 * 2)^{\sum_{n_2=0}^{M-1} (\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M))} * \\
& (-1)^{(\sum_{n_2=0}^{M-1} (\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M)) + 3) / 2} * \\
& \sum_{n_2=0}^{M-1} ((2 * n_2 + 1) * (\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M))) * \\
& n_0 \\
& (n_0 + 1)^{-x_1} * \\
& (n_0 + 2)^{-\sum_{n_2=0}^{M-1} ((2 * n_2 + 1) * (\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M)))} * \\
& \Pi_{n_2=0}^{M-1} (\text{factorial}(\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M))^{-1} * (2 * n_2 + 1)^{-\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M)}) \\
&)) \\
& - (\sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + \sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(\text{floor}(M) / \text{abs}(x_2) + q_2 / x_2) * 2\pi * (1 - x_1)} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) \\
&))))
\end{aligned}$$

lihtsustan viies n0-summa n1-summa sisse. ja muudan summade indekseid.

$$\begin{aligned}
& \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} (\sum_{n_0=0}^{(2 * M)^M - 1} (\\
& (\sum_{n_1=0}^{M-1} (1 + \text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M)) \% 2) * \\
& (x_2 * 2)^{\sum_{n_1=0}^{M-1} (\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M))} * \\
& (-1)^{\sum_{n_1=0}^{M-1} (\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M)) / 2} * \\
& \sum_{n_1=0}^{\text{floor}(e^{(\text{floor}(M) / \text{abs}(x_2) + q_1 / x_2) * 2\pi}) - 1} (\\
& \sum_{n_2=0}^{M-1} ((2 * n_2 + 1) * (\text{floor}(n_0 / (2 * M)^{n_2}) \% (2 * M))) * \\
& n_1 \\
& (n_1 + 1)^{-x_1} * \\
& (n_1 + 2)^{-\sum_{n_2=0}^{M-1} ((2 * n_2 + 1) * (\text{floor}(n_0 / (2 * M)^{n_2}) \% (2 * M)))} * \\
&) \\
& \Pi_{n_1=0}^{M-1} (\text{factorial}(\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M))^{-1} * (2 * n_1 + 1)^{-\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M)}) \\
&) \\
& - (\sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + \sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(\text{floor}(M) / \text{abs}(x_2) + q_1 / x_2) * 2\pi * (1 - x_1)} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n_0=0}^{(2 * M)^M - 1} (\\
& (\sum_{n_1=0}^{M-1} (\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M)) \% 2) * \\
& (x_2 * 2)^{\sum_{n_1=0}^{M-1} (\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M))} * \\
& (-1)^{(\sum_{n_1=0}^{M-1} (\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M)) + 3) / 2} * \\
& \sum_{n_1=0}^{\text{floor}(e^{(\text{floor}(M) / \text{abs}(x_2) + q_2 / x_2) * 2\pi}) - 1} (\\
& \sum_{n_2=2}^{M-1} ((2 * n_2 + 1) * (\text{floor}(n_0 / (2 * M)^{n_2}) \% (2 * M))) * \\
& n_1 \\
& (n_1 + 1)^{-x_1} * \\
& (n_1 + 2)^{-\sum_{n_2=0}^{M-1} ((2 * n_2 + 1) * (\text{floor}(n_0 / (2 * M)^{n_2}) \% (2 * M)))} * \\
&) \\
& \Pi_{n_1=0}^{M-1} (\text{factorial}(\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M))^{-1} * (2 * n_1 + 1)^{-\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M)}) \\
&) \\
& - (\sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + \sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(\text{floor}(M) / \text{abs}(x_2) + q_2 / x_2) * 2\pi * (1 - x_1)} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) \\
&))))
\end{aligned}$$

lihtsustan n1-summa(vana n0-summa) sees olevat osa:

$$\begin{aligned}
& \forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} (\sum_{n_0=0}^{(2 * M)^M - 1} (
\end{aligned}$$

$$\begin{aligned}
& (\sum_{n_1=0}^{M-1} (1 + \text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M)) \% 2) * \\
& (x_2 * 2)^{\sum_{n_1=0}^{M-1} (\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M))} * \\
& (-1)^{\sum_{n_1=0}^{M-1} (\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M)) / 2} * \\
& \sum_{n_1=2}^{\text{floor}(e^{(\text{floor}(M) / \text{abs}(x_2) + q_1 / x_2) * 2\pi}) + 1} (\\
& (1 - 2 / n_1)^{\sum_{n_2=0}^{M-1} ((2 * n_2 + 1) * (\text{floor}(n_0 / (2 * M)^{n_2}) \% (2 * M)))} * \\
& (n_1 - 1)^{-x_1} * \\
&) \\
& \prod_{n_1=0}^{M-1} (\text{factorial}(\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M))^{-1} * (2 * n_1 + 1)^{-\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M)}) \\
&) \\
& - (\sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + \sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(\text{floor}(M) / \text{abs}(x_2) + q_1 / x_2) * 2\pi * (1 - x_1)} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n_0=0}^{(2 * M)^{M-1}} (\\
& (\sum_{n_1=0}^{M-1} (\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M)) \% 2) * \\
& (x_2 * 2)^{\sum_{n_1=0}^{M-1} (\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M))} * \\
& (-1)^{(\sum_{n_1=0}^{M-1} (\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M)) + 3) / 2} * \\
& \sum_{n_1=2}^{\text{floor}(e^{(\text{floor}(M) / \text{abs}(x_2) + q_2 / x_2) * 2\pi}) + 1} (\\
& (1 - 2 / n_1)^{\sum_{n_2=2}^{M-1} ((2 * n_2 + 1) * (\text{floor}(n_0 / (2 * M)^{n_2}) \% (2 * M)))} * \\
& (n_1 - 1)^{-x_1} * \\
&) \\
& \prod_{n_1=0}^{M-1} (\text{factorial}(\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M))^{-1} * (2 * n_1 + 1)^{-\text{floor}(n_0 / (2 * M)^{n_1}) \% (2 * M)}) \\
&) \\
& - (\sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + \sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(\text{floor}(M) / \text{abs}(x_2) + q_2 / x_2) * 2\pi * (1 - x_1)} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) \\
&))))
\end{aligned}$$

1.2.3 tagumine pool nulliks

$$\begin{aligned}
& \sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + \sin((q_1 + 0/4) * 2\pi) * x_2 = 0 \wedge \\
& \sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + \sin((q_2 + 1/4) * 2\pi) * x_2 = 0
\end{aligned}$$

$$\begin{aligned}
& \sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + \sin(q_1 * 2\pi) * x_2 = 0 \wedge \\
& \sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + \sin((q_2 + 1/4) * 2\pi) * x_2 = 0
\end{aligned}$$

$$\begin{aligned}
& \exists_k (k \in N \wedge q_1 = k/2 - \arctan((1 - x_1) / x_2) / (2\pi)) \\
& \exists_k (k \in N \wedge q_2 = k/2 + \arctan(x_2 / (1 - x_1)) / (2\pi))
\end{aligned}$$

$$\exists_k (k \in N \wedge q_2 = k/2 - \arctan((1 - x_1) / x_2) / (2\pi) + 1/4)$$

$$\begin{aligned}
& \arctan(x_2 / (1 - x_1)) + \arctan((1 - x_1) / x_2) = 2\pi/4 * \text{sgn}((1 - x_1) * x_2) \\
& \arctan(x_2 / (1 - x_1)) = -\arctan((1 - x_1) / x_2) + 2\pi/4 * \text{sgn}((1 - x_1) * x_2)
\end{aligned}$$

$$\exists_k (k \in N \wedge q_2 = k/2 - \arctan((1 - x_1) / x_2) / (2\pi) + 1/4 * \text{sgn}((1 - x_1) * x_2))$$

$$\begin{aligned}
& q_2 = q_1 + 1/4 * \text{sgn}((1 - x_1) * x_2) + k/2 \\
& q_2 = q_1 + 1/4 \\
& \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) =
\end{aligned}$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} \left(\sum_{n=1}^{\text{floor}(e^{\text{floor}(M)/\text{abs}(x_2)} + (k/2 - \arctan((1-x_1)/x_2)/(2\pi))/x_2) * 2\pi} \right) (\sin((x_2 * \ln(n)/(2\pi) + 1/4) * 2\pi) * n^{-x_1}) \\
& + i * \lim_{M \rightarrow \infty} \left(- \sum_{n=1}^{\text{floor}(e^{\text{floor}(M)/\text{abs}(x_2)} + (k/2 + \arctan(x_2/(1-x_1))/(2\pi))/x_2) * 2\pi} \right) (\sin((x_2 * \ln(n)/(2\pi)) * 2\pi) * n^{-x_1}) = \\
& \lim_{M \rightarrow \infty} \left(\sum_{n=1}^{\text{floor}(e^{\text{floor}(M) * 2\pi/\text{abs}(x_2)} + k * 2\pi/2 - \arctan((1-x_1)/x_2)/x_2)} \right) (\sin(x_2 * \ln(n) + 2\pi/4) * n^{-x_1}) \\
& + i * \lim_{M \rightarrow \infty} \left(- \sum_{n=1}^{\text{floor}(e^{\text{floor}(M) * 2\pi/\text{abs}(x_2)} + k * 2\pi/2 + \arctan(x_2/(1-x_1))/x_2)} \right) (\sin(x_2 * \ln(n)) * n^{-x_1}) = \\
& \lim_{M \rightarrow \infty} \left(\sum_{n=1}^{\text{floor}(e^{\text{floor}(M) * 2\pi/2/\text{abs}(x_2)} - \arctan((1-x_1)/x_2)/x_2)} \right) (\sin(x_2 * \ln(n) + 2\pi/4) * n^{-x_1}) \\
& + i * \lim_{M \rightarrow \infty} \left(- \sum_{n=1}^{\text{floor}(e^{\text{floor}(M) * 2\pi/2/\text{abs}(x_2)} + \arctan(x_2/(1-x_1))/x_2)} \right) (\sin(x_2 * \ln(n)) * n^{-x_1}) = \\
& \lim_{M \rightarrow \infty} \left(\sum_{n=1}^{\text{floor}(e^{\text{floor}(M) * 2\pi/2/\text{abs}(x_2)} - \arctan((1-x_1)/x_2)/x_2)} \right) (\sin(x_2 * \ln(n) + 2\pi/4) * n^{-x_1}) \\
& + i * \lim_{M \rightarrow \infty} \left(- \sum_{n=1}^{\text{floor}(e^{\text{floor}(M) * 2\pi/2/\text{abs}(x_2)} + (-\arctan((1-x_1)/x_2)/x_2 + 2\pi/4 * \text{sgn}((1-x_1)*x_2))/x_2)} \right) (\sin(x_2 * \ln(n)) * n^{-x_1}) = \\
& \lim_{M \rightarrow \infty} \left(\sum_{n=1}^{\text{floor}(e^{\text{floor}(M) * 2\pi/2/\text{abs}(x_2)} - \arctan((1-x_1)/x_2)/x_2)} \right) (\sin(x_2 * \ln(n) + 2\pi/4) * n^{-x_1}) \\
& + i * \lim_{M \rightarrow \infty} \left(- \sum_{n=1}^{\text{floor}(e^{\text{floor}(M) * 2\pi/2/\text{abs}(x_2)} - \arctan((1-x_1)/x_2)/x_2 + 2\pi/4 * \text{sgn}((1-x_1)*x_2)/x_2)} \right) (\sin(x_2 * \ln(n)) * n^{-x_1}) =
\end{aligned}$$

))

Valem zeta13_tagumine_pool_nulliks:

$$\begin{aligned}
& \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} \left(\sum_{n=1}^{\text{floor}(e^{\text{floor}(M)/2 * 2\pi/\text{abs}(x_2)} - \arctan((1-x_1)/x_2)/x_2)} \right) (\sin(x_2 * \ln(n) + 2\pi/4) * n^{-x_1}) \\
& + i * \lim_{M \rightarrow \infty} \left(- \sum_{n=1}^{\text{floor}(e^{\text{floor}(M)/2 + 1/4} * 2\pi/\text{abs}(x_2) - \arctan((1-x_1)/x_2)/x_2)} \right) (\sin(x_2 * \ln(n)) * n^{-x_1}) \\
&))
\end{aligned}$$

asendan siinuse ja logaritmi vastavate Tayloriga.

$$\begin{aligned}
& \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} \left(\sum_{n_0=1}^{\text{floor}(e^{\text{floor}(M)/2 * 2\pi/\text{abs}(x_2)} - \arctan((1-x_1)/x_2)/x_2)} \right) (\\
& \sum_{n_1=0}^M (\\
& (-1)^{n_1} * (\text{fact}(2 * n_1))^{-1} * (x_2 * \ln(n_0) + 2\pi/4)^{2 * n_1} * n_0^{-x_1}))) \\
& + i * \lim_{M \rightarrow \infty} \left(- \sum_{n_0=1}^{\text{floor}(e^{\text{floor}(M)/2 + 1/4} * 2\pi/\text{abs}(x_2) - \arctan((1-x_1)/x_2)/x_2)} \right) (\sin(x_2 * \ln(n_0)) * n_0^{-x_1}) \\
&))
\end{aligned}$$

asendan logaritmi vastavate Tayloriga.

valem zeta14_tagumine_pool_nulliks_tayloriga(ei suutnud kontrollida, aga vb sest arvutusvõime piiratud tõttu ei saanud piidsavaõlt suurt Mi vsalida):

$$\begin{aligned}
& \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) = \\
& \lim_{M \rightarrow \infty} \left(\sum_{n_0=0}^{\text{floor}(e^{\text{floor}(M)/2 * 2\pi/\text{abs}(x_2)} - \arctan((1-x_1)/x_2)/x_2)} \right) (\sum_{n_1=0}^{(2 * M)^M - 1} (\\
& (1 + \% (2 * M)) \% 2) * \\
& (x_2 * 2)^{\sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2 * M)^{n_2}) \% (2 * M))} * \\
& (-1)^{\sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2 * M)^{n_2}) \% (2 * M))/2} * \\
& \sum_{n_2=0}^{M-1} ((2 * n_2 + 1) * (\text{floor}(n_1/(2 * M)^{n_2}) \% (2 * M))) * \\
& n_0^{(n_0 + 1) - x_1} * \\
& (n_0 + 2)^{-\sum_{n_2=0}^{M-1} ((2 * n_2 + 1) * (\text{floor}(n_1/(2 * M)^{n_2}) \% (2 * M)))} * \\
& \prod_{n_2=0}^{M-1} (\text{factorial}(\text{floor}(n_1/(2 * M)^{n_2}) \% (2 * M))^{-1} * (2 * n_2 + 1)^{-\text{floor}(n_1/(2 * M)^{n_2}) \% (2 * M))} \\
&)))
\end{aligned}$$

$$\begin{aligned}
& + i * \lim_{M \rightarrow \infty} \left(\sum_{n_0=0}^{\text{floor}(e^{\text{floor}(M)/2 + 1/4} * 2\pi/\text{abs}(x_2) - \arctan((1-x_1)/x_2)/x_2)} \right) (\sum_{n_1=0}^{(2 * M)^M - 1} (\\
& \sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2 * M)^{n_2}) \% (2 * M)) \% 2) *
\end{aligned}$$

$$\begin{aligned}
& (x_2 * 2)^{\sum_{n_2=0}^{M-1} (\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M))} * \\
& (-1)^{(\sum_{n_2=0}^{M-1} (\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M)) + 1) / 2} * \\
& \sum_{n_2=0}^{M-1} ((2 * n_2 + 1) * (\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M))) \\
& n_0 \\
& (n_0 + 1)^{-x_1} * \\
& (n_0 + 2)^{-\sum_{n_2=0}^{M-1} ((2 * n_2 + 1) * (\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M)))} * \\
& \prod_{n_2=0}^{M-1} (\text{factorial}(\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M))^{-1} * (2 * n_2 + 1)^{-\text{floor}(n_1 / (2 * M)^{n_2}) \% (2 * M)}) \\
&))) \\
&))
\end{aligned}$$

1.2.4 $q_1 = q_2$

$$\forall_{q_1} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M) / \text{abs}(x_2) + q / x_2) * 2\pi})} (\sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1}) \\
& - (\sin((q + 1/4) * 2\pi) * (1 - x_1) + \sin((q + 0/4) * 2\pi) * x_2) * e^{(\text{floor}(M) / \text{abs}(x_2) + q / x_2) * 2\pi * (1 - x_1) / x_2} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M) / \text{abs}(x_2) + q / x_2) * 2\pi})} (\sin(x_2 * \ln(n) + (2\pi) * 2/4) * n^{-x_1}) \\
& - (\sin((q + 2/4) * 2\pi) * (1 - x_1) + \sin((q + 1/4) * 2\pi) * x_2) * e^{(\text{floor}(M) / \text{abs}(x_2) + q / x_2) * 2\pi * (1 - x_1) / x_2} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) \\
&))))
\end{aligned}$$

$q_1 = 0 \wedge q_2 = 0$ kui valida, et $q_1 = 0$ ja $q_2 = 0$. Valem zeta12:

$$\begin{aligned}
& \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) =
\end{aligned}$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M) * 2\pi / \text{abs}(x_2))})} (\sin(x_2 * \ln(n) + 1 * (2\pi) / 4) * n^{-x_1}) \\
& - e^{\text{floor}(M) * 2\pi * (1 - x_1) / \text{abs}(x_2)} * ((1 - x_1)^2 + x_2^2)^{-1} * (1 - x_1) \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M) * 2\pi / \text{abs}(x_2))})} (\sin(x_2 * \ln(n) + 2 * (2\pi) / 4) * n^{-x_1}) \\
& - e^{\text{floor}(M) * 2\pi * (1 - x_1) / \text{abs}(x_2)} * ((1 - x_1)^2 + x_2^2)^{-1} * x_2 \\
&) = \\
&))
\end{aligned}$$

1.2.5 $q_1 = 1/8 \wedge q_2 = -1/8$

$$\begin{aligned}
& \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\
& \zeta(x_1 + i * x_2) =
\end{aligned}$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M) / \text{abs}(x_2) + 1/8 / x_2) * 2\pi})} (\sin((x_2 * \ln(n) / (2\pi) + 1/4) * 2\pi) * n^{-x_1}) \\
& - (\sin((1/8 + 1/4) * 2\pi) * (1 - x_1) + \sin((1/8 + 0/4) * 2\pi) * x_2) * e^{(\text{floor}(M) / \text{abs}(x_2) + 1/8 / x_2) * 2\pi * (1 - x_1)} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M) / \text{abs}(x_2) - 1/8 / x_2) * 2\pi})} (\sin((x_2 * \ln(n) / (2\pi) + 2/4) * 2\pi) * n^{-x_1}) \\
& - (\sin((-1/8 + 2/4) * 2\pi) * (1 - x_1) + \sin((-1/8 + 1/4) * 2\pi) * x_2) * e^{(\text{floor}(M) / \text{abs}(x_2) - 1/8 / x_2) * 2\pi * (1 - x_1) / x_2} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&)
\end{aligned}$$

$$x_2^2)^{-1} \\) =$$

$$\begin{aligned} & \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+1/8/x_2)*2\pi})} (\sin((x_2 * \ln(n))/(2\pi) + 1/4) * 2\pi) * n^{-x_1} \\ & - ((1-x_1) + x_2) * e^{(\text{floor}(M)/\text{abs}(x_2)+1/8/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} * 2^{-1/2} \\ &) \\ & + i * \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)-1/8/x_2)*2\pi})} (\sin((x_2 * \ln(n))/(2\pi) + 2/4) * 2\pi) * n^{-x_1} \\ & - ((1-x_1) + x_2) * e^{(\text{floor}(M)/\text{abs}(x_2)-1/8/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} * 2^{-1/2} \\ &) = \\ &)) \end{aligned}$$

1.2.6 kirjutades siinuse ja koosiinuse summana

$$\begin{aligned} \text{kasutatud valem} \quad \sin(x) &= \sum_{m=0}^{\infty} (x^{2*m+1} * ((2*m+1)!)^{-1} * (-1)^m) \\ \cos(x) &= \sum_{m=0}^{\infty} (x^{2*m} * ((2*m)!)^{-1} * (-1)^m) \end{aligned}$$

$$\begin{aligned} \text{zeta ise} \quad \forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow \\ \zeta(x_1 + i * x_2) &= \\ (x_1^2 - 2 * x_1 + 1 + x_2^2)^{-1} * \sum_{n=1}^{\infty} (\\ & \sum_{m=0}^{\infty} (\ln(n * (2 * m + 1))) * x_2^{2*m+1} * ((2 * m + 1)!)^{-1} * (-1)^m * x_2 * (n-1) * n^{-x_1} + \\ & \sum_{m=0}^{\infty} (\ln(n * 2 * m)) * x_2^{2*m} * ((2 * m)!)^{-1} * (-1)^m * (x_1^2 - x_1 - n * x_1 + n + x_2^2) * n^{-x_1} - \\ & \sum_{m=0}^{\infty} (\ln((n+1) * (2 * m + 1))) * x_2^{2*m+1} * ((2 * m + 1)!)^{-1} * (-1)^m * x_2 * n * (n+1)^{-x_1} - \\ & \sum_{m=0}^{\infty} (\ln((n+1) * 2 * m)) * x_2^{2*m} * ((2 * m)!)^{-1} * (-1)^m * (1-x_1) * n * (n+1)^{-x_1} \\ &) \\ & + i * (x_1^2 - 2 * x_1 + 1 + x_2^2)^{-1} * \sum_{n=1}^{\infty} (\\ & (\sin(x_2 * \ln(n) + 1 * (2\pi)/4) * x_2 * (n-1) + \sin(x_2 * \ln(n) + 2 * (2\pi)/4) * (x_1^2 - x_1 - n * x_1 + n + x_2^2)) * n^{-x_1} - \\ & (\sin(x_2 * \ln(n+1) + 1 * (2\pi)/4) * x_2 + \sin(x_2 * \ln(n+1) + 2 * (2\pi)/4) * (1-x_1)) * n * (n+1)^{-x_1} \\ &) \\ & = \\ & (x_1^2 - 2 * x_1 + 1 + x_2^2)^{-1} * \sum_{n=1}^{\infty} (\sum_{m=0}^{\infty} (x_2^{2*m} * (-1)^m * ((2 * m)!)^{-1} * (\\ & \ln(n * (2 * m + 1))) * (2 * m + 1)^{-1} * x_2^2 * (n-1) * n^{-x_1} + \\ & \ln(n * 2 * m) * (x_1^2 - x_1 - n * x_1 + n + x_2^2) * n^{-x_1} - \\ & \ln((n+1) * (2 * m + 1))) * (2 * m + 1)^{-1} * x_2^2 * n * (n+1)^{-x_1} - \\ & \ln((n+1) * 2 * m) * (1-x_1) * n * (n+1)^{-x_1} \\ &))) \\ & + i * (x_1^2 - 2 * x_1 + 1 + x_2^2)^{-1} * \sum_{n=1}^{\infty} (\\ & (\sin(x_2 * \ln(n) + 1 * (2\pi)/4) * x_2 * (n-1) + \sin(x_2 * \ln(n) + 2 * (2\pi)/4) * (x_1^2 - x_1 - n * x_1 + n + x_2^2)) * n^{-x_1} - \\ & (\sin(x_2 * \ln(n+1) + 1 * (2\pi)/4) * x_2 + \sin(x_2 * \ln(n+1) + 2 * (2\pi)/4) * (1-x_1)) * n * (n+1)^{-x_1} \\ &) \\ & =) \end{aligned}$$

2 zetafunktsiooni nullkohad

Võrdsustan zetafunktsiooni kuju zeta_2_q_ga, mille mujal tuletasin 0iga.

$$\forall_{q_1} (\forall_{q_2} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$

$$\begin{aligned} & \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1}) \\ & - (\sin((q_1 + 1/4) * 2\pi) * (1-x_1) + \sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\ &) \end{aligned}$$

$$\begin{aligned}
&) \\
& + i * \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_2)*2\pi/x_2})} (\sin(x_2 * \ln(n) + (2\pi) * 2/4) * n^{-x_1}) \\
& - (\sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + \sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(\text{floor}(M)+q_2)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) = 0 \\
&))))
\end{aligned}$$

kuna nii reaalsosa kui imaginaarsosa peavad nullid olema, siis võin imaginnaarsosa suvalise konstandiga läbi korrutada.

$$\forall_{q_1} (\forall_{q_2} (\forall_{Q_1} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$

$$\begin{aligned}
& Q_1 * \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1}) \\
& - (\sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + \sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} = 0 \\
&) + \\
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_2)*2\pi/x_2})} (\sin(x_2 * \ln(n) + (2\pi) * 2/4) * n^{-x_1}) \\
& - (\sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + \sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(\text{floor}(M)+q_2)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) = 0 \\
&))))
\end{aligned}$$

viin konsatndi sulguse sisse.

$$\forall_{q_1} (\forall_{q_2} (\forall_{Q_1} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& Q_1 * \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1}) \\
& - Q_1 * (\sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + \sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} + \\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_2)*2\pi/x_2})} (\sin(x_2 * \ln(n) + (2\pi) * 2/4) * n^{-x_1}) \\
& - (\sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + \sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(\text{floor}(M)+q_2)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} \\
&) = 0 \\
&))))
\end{aligned}$$

muudan kirjutan ümber viies ühiseid kordajaid sulguse ette.

$$\forall_{q_1} (\forall_{q_2} (\forall_{Q_1} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (Q_1 * \sin(x_2 * \ln(n) + (2\pi) * 1/4) * n^{-x_1} + \sin(x_2 * \ln(n) + (2\pi) * 2/4) * n^{-x_1}) \\
& - ((Q_1 * \sin((q_1 + 1/4) * 2\pi) + \sin((q_2 + 2/4) * 2\pi)) * (1 - x_1) + (Q_1 * \sin((q_1 + 0/4) * 2\pi) + \sin((q_2 + 1/4) * 2\pi)) * \\
& x_2) * e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1}) = 0 \\
&))))
\end{aligned}$$

siinuse ja koosinuse summa saab ühiseks siinuseks teha nii, et argumendile on midagi liidetud.

$$\forall_{q_1} (\forall_{Q_1} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sqrt{Q_1^2 + 1} * -\sin(x_2 * \ln(n) + \arctan(-Q_1)) * n^{-x_1}) \\
& - ((-\sqrt{Q_1^2 + 1} * \sin(q_1 * 2\pi + \arctan(-Q_1))) * (1 - x_1) + \sqrt{Q_1^2 + 1} * \sin(q_1 * 2\pi + \arctan(1/Q_1)) * x_2) \\
& e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1}) = 0 \\
&))))
\end{aligned}$$

$\arctan(-x) = \arctan(x)$

$$\forall_{q_1} (\forall_{Q_1} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (-\sin(x_2 * \ln(n) - \arctan(Q_1)) * n^{-x_1})
\end{aligned}$$

$$\begin{aligned}
& -(-\sin(q_1 * 2\pi - \arctan(Q_1)) * (1 - x_1) + \sin(q_1 * 2\pi + \arctan(1/Q_1)) * x_2) \\
& e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}) \\
&) = 0 \\
&)))
\end{aligned}$$

$$\arctan(1/x) = -\arctan(x) + \text{sgn}(x) * 2\pi/4$$

$$\forall_{q_1} (\forall_{Q_1} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (-\sin(x_2 * \ln(n) - \arctan(Q_1)) * n^{-x_1}) \\
& -(-\sin(q_1 * 2\pi - \arctan(Q_1)) * (1 - x_1) + \sin(q_1 * 2\pi - \arctan(Q_1) + \text{sgn}(Q_1) * 2\pi/4) * x_2) \\
& e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}) \\
&) = 0 \\
&)))
\end{aligned}$$

korrutan -1ega läbi

$$\forall_{q_1} (\forall_{Q_1} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) - \arctan(Q_1)) * n^{-x_1}) \\
& -(\sin(q_1 * 2\pi - \arctan(Q_1)) * (1 - x_1) - \sin(q_1 * 2\pi - \arctan(Q_1) + \text{sgn}(Q_1) * 2\pi/4) * x_2) \\
& e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}) \\
&) = 0 \\
&)))
\end{aligned}$$

2pi/4 lahutamise on sama kui 2pi/4 lahutamise ja siis -1'ega läbi korrutamine

$$\forall_{q_1} (\forall_{Q_1} (\forall_{x_1} (\forall_{x_2} (x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) - \arctan(Q_1)) * n^{-x_1}) \\
& -(\sin(q_1 * 2\pi - \arctan(Q_1)) * (1 - x_1) - \sin(q_1 * 2\pi - \arctan(Q_1) + 2\pi/4) * \text{sgn}(Q_1) * x_2) \\
& e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}) \\
&) = 0 \\
&)))
\end{aligned}$$

kui Q_1 on positiivne, siis $\text{sgn}(Q_1)=1$, kui negatiivne, siis $\text{sgn}(Q_1)=-1$

$$\forall_{q_1} (\forall_{x_1} (\forall_{x_2} (\forall_{Q_1} (q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow ($$

$Q_1 > 0$ and

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) - \arctan(Q_1)) * n^{-x_1}) \\
& -(\sin(q_1 * 2\pi - \arctan(Q_1)) * (1 - x_1) - \sin(q_1 * 2\pi - \arctan(Q_1) + 2\pi/4) * x_2) \\
& e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}) \\
&) = 0 \\
& \vee
\end{aligned}$$

$Q_1 < 0$ and

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) - \arctan(Q_1)) * n^{-x_1}) \\
& -(\sin(q_1 * 2\pi - \arctan(Q_1)) * (1 - x_1) + \sin(q_1 * 2\pi - \arctan(Q_1) + 2\pi/4) * x_2) \\
& e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}) \\
&) = 0 \\
&))))
\end{aligned}$$

$\arctan(Q_1)$ võtab väärtuseid $-2\pi/4$ kuni $2\pi/4$

$$\forall_{q_1} (\forall_{x_1} (\forall_{x_2} (\forall_{Q_1} (q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow ($$

$(Q_1 > -2\pi/4 \wedge Q_1 < 0) \wedge ($

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) - Q_1) * n^{-x_1})
\end{aligned}$$

$$- (\sin(q_1 * 2\pi - Q_1) * (1 - x_1) - \sin(q_1 * 2\pi - Q_1 + 2\pi/4) * x_2) \\ e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\) = 0 \vee$$

$$(Q_1 > 0 \wedge Q_1 < 2\pi/4) \wedge (\\ \lim_{M \rightarrow \infty} (\\ \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) - Q_1) * n^{-x_1}) \\ - (\sin(q_1 * 2\pi - Q_1) * (1 - x_1) + \sin(q_1 * 2\pi - Q_1 + 2\pi/4) * x_2) \\ e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\) = 0 \\)$$

))))

Ühes kohas Q_1 asemele samanimeline muutuja $-Q_1$. Teises kohas Q_1 asemele samanimeline muutuja $2\pi/2 - Q_1$.

$$\forall_{q_1} (\forall_{x_1} (\forall_{x_2} (\forall_{Q_1} (q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow (\\ (Q_1 > 0 \wedge Q_1 < 2\pi/4) \wedge (\\ \lim_{M \rightarrow \infty} (\\ \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1) * n^{-x_1}) \\ - (\sin(q_1 * 2\pi + Q_1) * (1 - x_1) - \sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2) \\ e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\) = 0 \vee$$

$$(Q_1 > 2\pi/4 \wedge Q_1 < 2\pi/2) \wedge (\\ \lim_{M \rightarrow \infty} (\\ \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) - 2\pi/2 + Q_1) * n^{-x_1}) \\ - (\sin(q_1 * 2\pi - 2\pi/2 + Q_1) * (1 - x_1) + \sin(q_1 * 2\pi - 2\pi/2 + Q_1 + 2\pi/4) * x_2) \\ e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\) = 0 \\)$$

))))

arvestan, et $\sin(x-2\pi/2)=-\sin(x)$

$$\forall_{q_1} (\forall_{x_1} (\forall_{x_2} (\forall_{Q_1} (q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow (\\ (Q_1 > 0 \wedge Q_1 < 2\pi/4) \wedge (\\ \lim_{M \rightarrow \infty} (\\ \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1) * n^{-x_1}) \\ - (\sin(q_1 * 2\pi + Q_1) * (1 - x_1) - \sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2) \\ e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\) = 0 \vee$$

$$(Q_1 > 2\pi/4 \wedge Q_1 < 2\pi/2) \wedge (\\ \lim_{M \rightarrow \infty} (\\ \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (-\sin(x_2 * \ln(n) + Q_1) * n^{-x_1}) \\ - (-\sin(q_1 * 2\pi + Q_1) * (1 - x_1) - \sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2) \\ e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\) = 0 \\)$$

))))

Arvestan, et $0*(-1)=0$ ja $\sin(x+2\pi/2)=-\sin(x)$. Muutuja Q_1 asemele muutuja $Q_1-2\pi/2$.

$$\forall_{q_1} (\forall_{x_1} (\forall_{x_2} (\forall_{Q_1} (q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow ($$

$$\begin{aligned}
& (Q_1 > 0 \wedge Q_1 < 2\pi/4) \wedge (\\
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{f\text{loor}(e^{(f\text{loor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1) * n^{-x_1}) \\
& - (\sin(q_1 * 2\pi + Q_1) * (1 - x_1) - \sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2) \\
& e^{(f\text{loor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) = 0 \\
& \vee
\end{aligned}$$

$$\begin{aligned}
& (Q_1 > 2\pi/4 \wedge Q_1 < 2\pi/2) \wedge (\\
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{f\text{loor}(e^{(f\text{loor}(M)+q_1)*2\pi/x_2})} (-\sin(x_2 * \ln(n) + Q_1) * n^{-x_1}) \\
& - (-\sin(q_1 * 2\pi + Q_1) * (1 - x_1) - \sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2) \\
& e^{(f\text{loor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) = 0 \\
& \vee
\end{aligned}$$

$$\begin{aligned}
& (Q_1 > 2\pi/2 \wedge Q_1 < 2\pi * 3/4) \wedge (\\
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{f\text{loor}(e^{(f\text{loor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1) * n^{-x_1}) \\
& - (\sin(q_1 * 2\pi + Q_1) * (1 - x_1) - \sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2) \\
& e^{(f\text{loor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) = 0 \\
& \vee
\end{aligned}$$

$$\begin{aligned}
& (Q_1 > 2\pi * 3/4 \wedge Q_1 < 2\pi) \wedge (\\
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{f\text{loor}(e^{(f\text{loor}(M)+q_1)*2\pi/x_2})} (-\sin(x_2 * \ln(n) + Q_1) * n^{-x_1}) \\
& - (-\sin(q_1 * 2\pi + Q_1) * (1 - x_1) - \sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2) \\
& e^{(f\text{loor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) = 0
\end{aligned}$$

)

)))

korrutan osad osad -lega läbi. ja panaen osad kokku.

$$\begin{aligned}
& \forall_{q_1} (\forall_{x_1} (\forall_{x_2} (\forall_{Q_1} (q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow (\\
& (Q_1 > 0 \wedge Q_1 < 2\pi/4 \vee Q_1 > 2\pi/2 \wedge Q_1 < 2\pi * 3/4) \wedge (\\
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{f\text{loor}(e^{(f\text{loor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1) * n^{-x_1}) \\
& - (\sin(q_1 * 2\pi + Q_1) * (1 - x_1) - \sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2) \\
& e^{(f\text{loor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) = 0 \\
& \vee
\end{aligned}$$

$$\begin{aligned}
& (Q_1 > 2\pi/4 \wedge Q_1 < 2\pi/2 \vee Q_1 > 2\pi * 3/4 \wedge Q_1 < 2\pi) \wedge (\\
& \lim_{M \rightarrow \infty} (\\
& \sum_{n=1}^{f\text{loor}(e^{(f\text{loor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1) * n^{-x_1}) \\
& - (\sin(q_1 * 2\pi + Q_1) * (1 - x_1) + \sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2) \\
& e^{(f\text{loor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \\
&) = 0
\end{aligned}$$

)

))))

üheks oskas kokku.

$$\begin{aligned} & \forall_{q_1} (\forall_{x_1} (\forall_{x_2} (\forall_{Q_1} (q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow (\\ & \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1) * n^{-x_1}) \\ & - (\sin(q_1 * 2\pi + Q_1) * (1 - x_1) - \sin(q_1 * 2\pi + Q_1 + 2\pi/4) * \text{sgn}(Q_1 \% (2\pi/2) - 2\pi/4) * x_2) \\ & e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) = 0 \\ &) \\ &)))) \end{aligned}$$

muutuja Q_1 asemele Q_1/(2\pi)

$$\begin{aligned} & \forall_{q_1} (\forall_{x_1} (\forall_{x_2} (\forall_{Q_1} (q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow (\\ & \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1 * 2\pi) * n^{-x_1}) \\ & - (\sin((q_1 + Q_1) * 2\pi) * (1 - x_1) - \sin((q_1 + Q_1 + 1/4) * 2\pi) * ((\text{floor}(Q_1 * 4) \% 2) * 2 - 1) * x_2) \\ & e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) = 0 \\ &) \\ &)))) \end{aligned}$$

kuna Q_1'ele 2\pi liites...

$$\begin{aligned} & \forall_{q_1} (\forall_{x_1} (\forall_{x_2} (\forall_{Q_1} (q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow (\\ & \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1 * 2\pi) * n^{-x_1}) \\ & - (\sin((q_1 + Q_1) * 2\pi) * (1 - x_1) - \sin((q_1 + Q_1 + 1/4) * 2\pi) * ((\text{floor}(Q_1 * 4) \% 2) * 2 - 1) * x_2) \\ & e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) = 0 \\ & \wedge \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1 * 2\pi) * n^{-x_1}) \\ & - (\sin((q_1 + Q_1) * 2\pi) * (1 - x_1) + \sin((q_1 + Q_1 + 1/4) * 2\pi) * ((\text{floor}(Q_1 * 4) \% 2) * 2 - 1) * x_2) \\ & e^{(\text{floor}(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) = 0 \\ &) \\ &)))) \end{aligned}$$

et see mõlemal juhul 0 oleks sin((q_1+Q_1+1/4)*2\pi)=0 seega q_1=-1/4-Q_1 või q_1=1/4-Q_1

$$\forall_{x_1} (\forall_{x_2} (\forall_{Q_1} (Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow ($$

$$\begin{aligned} & \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)-1/4-Q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1 * 2\pi) * n^{-x_1}) \\ & + (1 - x_1) * e^{(\text{floor}(M)-1/4-Q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) = 0 \\ & \vee \lim_{M \rightarrow \infty} (\\ & \sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)+1/4-Q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1 * 2\pi) * n^{-x_1}) \\ & - (1 - x_1) * e^{(\text{floor}(M)+1/4-Q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} \\ &) = 0 \\ &) \\ &))) \end{aligned}$$

samahästi võib m'i 2 korda aeglasemalt suurendada.

$$\forall_{x_1} (\forall_{x_2} (\forall_{Q_1} (Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow ($$

$$\begin{aligned}
& \lim_{M \rightarrow \infty} \left(\sum_{n=1}^{\text{floor}(e^{(\text{floor}(M)/2+1/4-Q_1)*2\pi/x_2})} (\sin(x_2 * \ln(n) + Q_1 * 2\pi) * n^{-x_1}) \right. \\
& \left. - (1-x_1) * ((\text{floor}(M)\%2) * 2 - 1) * e^{(\text{floor}(M)/2+1/4-Q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1} \right) \\
& \left. \right) = 0 \\
& \left. \right) \\
& \left. \right)
\end{aligned}$$

3 muu

3.0.1 c

$$\begin{aligned}
& \lim_{M \rightarrow \infty} (M * (M+1)^{-x_1} - M^{1-x_1}) = \\
& x_1 \geq 0 \rightarrow \text{return}(0) \\
& x_1 < 0 \rightarrow \text{return}(\infty)
\end{aligned}$$

3.0.2 b

$$\begin{aligned}
& \lim_{M \rightarrow \infty} \left((\sin(x_2 * \ln(M+1) + 2 * (2\pi)/4) * (1-x_1) + \sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * x_2) * M * (M+1)^{-x_1} * (x_1^2 - 2 * x_1 + x_2^2 + 1)^{-1} - \right. \\
& \left. (\sin(x_2 * \ln(M+1) + 2 * (2\pi)/4) * (1-x_1) + \sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * x_2) * M^{1-x_1} * (x_1^2 - 2 * x_1 + x_2^2 + 1)^{-1} \right) = \\
& \lim_{M \rightarrow \infty} ((\sin(x_2 * \ln(M+1) + 2 * (2\pi)/4) * (1-x_1) + \sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * x_2) * \\
& (M * (M+1)^{-x_1} - M^{1-x_1})) \\
& \left. \right) = \\
& x_1 > 0 \rightarrow \text{return}(0) \\
& x_1 \leq 0 \rightarrow \text{return}(\infty)
\end{aligned}$$

3.0.3 a

$$\begin{aligned}
& \lim_{M \rightarrow \infty} \left((\sin(x_2 * \ln(M+1) + 2 * (2\pi)/4) * (1-x_1) + \sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * x_2) * M * (M+1)^{-x_1} * (x_1^2 - 2 * x_1 + x_2^2 + 1)^{-1} - \right. \\
& \left. (\sin(x_2 * \ln(M) + 2 * (2\pi)/4) * (1-x_1) + \sin(x_2 * \ln(M) + 1 * (2\pi)/4) * x_2) * M^{1-x_1} * (x_1^2 - 2 * x_1 + x_2^2 + 1)^{-1} \right) = \\
& \lim_{M \rightarrow \infty} \left((\sin(x_2 * \ln(M+1) + 2 * (2\pi)/4) * (1-x_1) + \sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * x_2) * M * (M+1)^{-x_1} - \right. \\
& \left. (\sin(x_2 * \ln(M) + 2 * (2\pi)/4) * (1-x_1) + \sin(x_2 * \ln(M) + 1 * (2\pi)/4) * x_2) * M^{1-x_1} \right) \\
& \left. \right) =
\end{aligned}$$

3.0.4 d

$$\begin{aligned}
& \lim_{M \rightarrow \infty} \left((\sin(x_2 * \ln(M+1) + 2 * (2\pi)/4) * (1-x_1) + \sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) * x_2) * M^{1-x_1} * (x_1^2 - 2 * x_1 + x_2^2 + 1)^{-1} - \right. \\
& \left. (\sin(x_2 * \ln(M) + 2 * (2\pi)/4) * (1-x_1) + \sin(x_2 * \ln(M) + 1 * (2\pi)/4) * x_2) * M^{1-x_1} * (x_1^2 - 2 * x_1 + x_2^2 + 1)^{-1} \right) = \\
& \left. \right) = \\
& (x_1^2 - 2 * x_1 + x_2^2 + 1)^{-1} * \lim_{M \rightarrow \infty} (M^{1-x_1} * (\\
& (1-x_1) * (\sin(x_2 * \ln(M+1) + 2 * (2\pi)/4) - \sin(x_2 * \ln(M) + 2 * (2\pi)/4)) + \\
& x_2 * (\sin(x_2 * \ln(M+1) + 1 * (2\pi)/4) - \sin(x_2 * \ln(M) + 1 * (2\pi)/4)) \\
& \left. \right)) =
\end{aligned}$$

$$(x_1^2 - 2 * x_1 + x_2^2 + 1)^{-1} * \lim_{M \rightarrow \infty} (M^{1-x_1} * ($$

$$(\cos(x_2 * \ln(M+1)) - \cos(x_2 * \ln(M))) * x_2 -$$

$$(\sin(x_2 * \ln(M+1)) - \sin(x_2 * \ln(M))) * (1 - x_1)))$$

$$\lim_{M \rightarrow \infty} (M^{1-x_1} * ($$

$$\text{abs}((\cos(x_2 * \ln(M+1)) - \cos(x_2 * \ln(M))) * x_2)) +$$

$$\text{abs}((\sin(x_2 * \ln(M+1)) - \sin(x_2 * \ln(M))) * (1 - x_1)))$$

$$))$$

$$\lim_{M \rightarrow \infty} (M^{1-x_1} * ($$

$$\text{abs}((x_2 * \ln(M+1) - x_2 * \ln(M)) * x_2) +$$

$$\text{abs}((x_2 * \ln(M+1) - x_2 * \ln(M)) * (1 - x_1)))$$

$$))$$

$$\lim_{M \rightarrow \infty} (M^{1-x_1} * ($$

$$\text{abs}((\ln(M+1) - \ln(M)) * x_2^2) +$$

$$\text{abs}((\ln(M+1) - \ln(M)) * (1 - x_1) * x_2)$$

$$))$$

$$\lim_{M \rightarrow \infty} (M^{1-x_1} * ($$

$$(\ln(M+1) - \ln(M)) * x_2^2 +$$

$$(\ln(M+1) - \ln(M)) * \text{abs}((1 - x_1) * x_2)$$

$$))$$

$$\lim_{M \rightarrow \infty} (M^{1-x_1} * ($$

$$(\ln(M+1) - \ln(M)) * (\text{abs}(x_2) * (\text{abs}(x_2) + \text{abs}(1 - x_1))))$$

$$))$$

$$\lim_{M \rightarrow \infty} (M^{1-x_1} * (\ln(M+1) - \ln(M)) * (\text{abs}(x_2) * (\text{abs}(x_2) + \text{abs}(1 - x_1))))$$

$$\lim_{M \rightarrow \infty} (M^{1-x_1} * (\ln(M+1) - \ln(M)))$$

$$\lim_{M \rightarrow \infty} (M^{-x_1} * (\ln(M+1) - \ln(M)) * M)$$

$$\lim_{M \rightarrow \infty} (M^{-x_1})$$

$$x_1 > 0 \rightarrow \text{return } 0$$

3.0.5 f

$$\ln(M) * x_2 = (k * 2\pi) / 2$$

$$M = e^{(k * 2\pi) / (2 * x_2)} = e^{\frac{k * 2\pi}{2 * x_2}}$$

$$k = 2 * \ln(M) * x_2 / (2\pi) = \frac{2 * \ln(M) * x_2}{2\pi}$$

$$x_2 = k * 2\pi / \ln(M) / 2 = \frac{k * 2\pi}{\ln(M) * 2}$$

asendus

$$M_{uus} = e^{(\text{floor}(M) + q) * 2\pi / x_2} = e^{\frac{\text{floor}(M) * 2\pi}{x_2}} = e^{(\text{floor}(M) * 2\pi + \alpha) / x_2}$$

asendus2

$$\ln(M) * x_2 = (k * 2\pi) + \alpha$$

$$M = e^{(k * 2\pi + \alpha) / (x_2)} = e^{\frac{k * 2\pi + \alpha}{x_2}}$$

$$k = (\ln(M) * x_2 - \alpha) / (2\pi) = \frac{\ln(M) * x_2 - \alpha}{2\pi}$$

$$x_2 = ((k * 2\pi) + \alpha) / \ln(M)$$

$$q = \alpha / 2\pi$$

3.0.6 g

$\sin(\ln(x)) =$
 $\forall_x (x > -1 \wedge x < 1 \rightarrow$
 $\ln(x+1) = \sum_{n=1}^{\infty} ((-1)^{n+1} * n^{-1} * x^n)$
 $)$
 muutuja vahetus x-1
 $\forall_x (x > 0 \wedge x < 2 \rightarrow$
 $\ln(x) = \sum_{n=1}^{\infty} ((-1)^{n+1} * n^{-1} * (x-1)^n)$
 $)$
 mv 1/x
 $\forall_x (x > 1/2 \rightarrow$
 $\ln(1/x) = \sum_{n=1}^{\infty} ((-1)^{n+1} * n^{-1} * (1/x - 1)^n)$
 $)$
 mv summas
 $\forall_x (x > 1/2 \rightarrow$
 $\ln(1/x) = \sum_{n=1}^{\infty} ((-1)^{n+1} * n^{-1} * (1/x - 1)^n)$
 $)$
 ..

3.0.7 ln maclaurin

<https://math.stackexchange.com/questions/585154/taylor-series-for-logx>

valem ln_maclaurin:
 $\forall_x (x > 0 \rightarrow$
 $\ln(x) = 2 * \sum_{k=0}^{\infty} ((2 * k + 1)^{-1} * (x - 1)^{2 * k + 1} * (x + 1)^{-2 * k - 1})$
 $)$

3.0.8 sin(ln(x)) maclaurin

sin:

$\forall_x (x \in R \rightarrow$
 $\sin(x) = \sum_{k=0}^{\infty} ((-1)^k * x^{2 * k + 1} * \Gamma(2 * k + 2)^{-1})$
 $)$
 sin(y*ln(x)): Valem sin_y_ln_x_maclaurin_3:
 $\forall_x (x > 0 \rightarrow$
 $\sin(y * \ln(x)) = \sum_{k_1=0}^{\infty} ((\sum_{k_2=0}^{\infty} ((2 * k_2 + 1)^{-1} * (x - 1)^{2 * k_2 + 1} * (x + 1)^{-2 * k_2 - 1}))^{2 * k_1 + 1} * y^{2 * k_1 + 1} * factorial(2 * k_1 + 1)^{-1} * 2^{2 * k_1 + 1} * (-1)^{k_1})$
 $)$

kasutan Multinomial theoreemi. Siin on eeldatud, et True'ga korrutamine on nagu 1ega korrutamine ja Falsega korrutamine on nagu 0'iga korrutamine. Valem sin_y_ln_x_multinomial_1:

$\forall_x (x > 0 \rightarrow$
 $\sin(y * \ln(x)) = \lim_{M \rightarrow \infty} (\sum_{k_1=0}^{M-1} (\sum_{k_2=0}^{(2 * k_1 + 2)^M - 1} ($
 $(\sum_{k_3=0}^{M-1} (floor(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)) = 2 * k_1 + 1) *$
 $\Pi_{k_3=0}^{M-1} (factorial(floor(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2))^{-1} * ((2 * k_3 + 1)^{-1} * (x - 1)^{2 * k_3 + 1} * (x + 1)^{-2 * k_3 - 1})^{floor(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)})$
 $) * y^{2 * k_1 + 1} * 2^{2 * k_1 + 1} * (-1)^{k_1})$
 $)$

elementide korrumtamise asemel võib nende astemed liita:

$$\begin{aligned} & \forall_x (x > 0 \rightarrow \sin(y * \ln(x)) = \\ & \lim_{M \rightarrow \infty} (\sum_{k_1=0}^{M-1} (\sum_{k_2=0}^{(2*k_1+2)^{M-1}} (\\ & (\sum_{k_3=0}^{M-1} (\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)) = 2 * k_1 + 1) * \\ & \Pi_{k_3=0}^{M-1} (\text{factorial}(\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2))^{-1} * (2 * k_3 + 1)^{-\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)}) * ((x - 1) / (x + \\ & 1))^{\sum_{k_3=0}^{M-1} ((2 * k_3 + 1) * (\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)))}) * y^{2 * k_1 + 1} * 2^{2 * k_1 + 1} * (-1)^{k_1})) \\ &) \end{aligned}$$

lihtsustan. Valem sin_y_ln_x_multinomial:

$$\begin{aligned} & \forall_x (x > 0 \rightarrow \sin(y * \ln(x)) = \\ & \lim_{M \rightarrow \infty} (\sum_{k_1=0}^{M-1} (y^{2 * k_1 + 1} * 2^{2 * k_1 + 1} * (-1)^{k_1} * \sum_{k_2=0}^{(2 * k_1 + 2)^{M-1}} (\\ & (\sum_{k_3=0}^{M-1} (\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)) = 2 * k_1 + 1) * \\ & (x - 1)^{\sum_{k_3=0}^{M-1} ((2 * k_3 + 1) * (\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)))} * \\ & (x + 1)^{-\sum_{k_3=0}^{M-1} ((2 * k_3 + 1) * (\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)))} * \\ & \Pi_{k_3=0}^{M-1} (\text{factorial}(\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2))^{-1} * (2 * k_3 + 1)^{-\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)}) \\ &)))) \\ &) \end{aligned}$$

viin teguri sulgude sisse::

$$\begin{aligned} & \forall_x (x > 0 \rightarrow \sin(y * \ln(x)) = \\ & \lim_{M \rightarrow \infty} (\sum_{k_1=0}^{M-1} (\sum_{k_2=0}^{(2 * k_1 + 2)^{M-1}} (\\ & (\sum_{k_3=0}^{M-1} (\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)) = 2 * k_1 + 1) * \\ & y^{2 * k_1 + 1} * 2^{2 * k_1 + 1} * (-1)^{k_1} * \\ & (x - 1)^{\sum_{k_3=0}^{M-1} ((2 * k_3 + 1) * (\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)))} * \\ & (x + 1)^{-\sum_{k_3=0}^{M-1} ((2 * k_3 + 1) * (\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)))} * \\ & \Pi_{k_3=0}^{M-1} (\text{factorial}(\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2))^{-1} * (2 * k_3 + 1)^{-\text{floor}(k_2 / (2 * k_1 + 2)^{k_3}) \% (2 * k_1 + 2)}) \\ &)))) \\ &) \end{aligned}$$

selle asemel et üle eri n2'e ristsummade summeerida arvutan lihtsalt n2'e ristsumma. Valem sin_y_ln_x_multinomial_2:

$$\begin{aligned} & \forall_x (x > 0 \rightarrow \sin(y * \ln(x)) = \\ & - \lim_{M \rightarrow \infty} (\sum_{k_1=0}^{(2 * M)^{M-1}} (\\ & (\sum_{k_2=0}^{M-1} (\text{floor}(k_1 / (2 * M)^{k_2}) \% (2 * M)) \% 2) * \\ & (\sum_{k_2=0}^{M-1} (\text{floor}(k_1 / (2 * M)^{k_2}) \% (2 * M)) < 2 * M) * \\ & (y * 2)^{\sum_{k_2=0}^{M-1} (\text{floor}(k_1 / (2 * M)^{k_2}) \% (2 * M))} * \\ & (-1)^{(\sum_{k_2=0}^{M-1} (\text{floor}(k_1 / (2 * M)^{k_2}) \% (2 * M)) - 1) / 2} * \\ & (x - 1)^{\sum_{k_2=0}^{M-1} ((2 * k_2 + 1) * (\text{floor}(k_1 / (2 * M)^{k_2}) \% (2 * M)))} * \\ & (x + 1)^{-\sum_{k_2=0}^{M-1} ((2 * k_2 + 1) * (\text{floor}(k_1 / (2 * M)^{k_2}) \% (2 * M)))} * \\ & \Pi_{k_2=0}^{M-1} (\text{factorial}(\text{floor}(k_1 / (2 * M)^{k_2}) \% (2 * M))^{-1} * (2 * k_2 + 1)^{-\text{floor}(k_1 / (2 * M)^{k_2}) \% (2 * M)}) \\ &)) \\ &) \end{aligned}$$

kuna Mi lähenedes lõpmatusle see avaldis nagu nii koondub ja selle muutuse tegemine ainult lisab liikmeid, mis on absoluutväärtuselt väiksemad kui oleksid liikmed, mis lisanduksid Mi suurendamisel. Valem sin_y_ln_x_multinomial_4:

$$\begin{aligned} & \forall_x (x > 0 \rightarrow \sin(y * \ln(x)) = \\ & \lim_{M \rightarrow \infty} (\sum_{k_1=0}^{(2 * M)^{M-1}} (\\ & (\sum_{k_2=0}^{M-1} (\text{floor}(k_1 / (2 * M)^{k_2}) \% (2 * M)) \% 2) * \\ & (y * 2)^{\sum_{k_2=0}^{M-1} (\text{floor}(k_1 / (2 * M)^{k_2}) \% (2 * M))} * \\ & (-1)^{(\sum_{k_2=0}^{M-1} (\text{floor}(k_1 / (2 * M)^{k_2}) \% (2 * M)) + 3) / 2} * \end{aligned}$$

$$\begin{aligned}
& (x-1)^{\sum_{k_2=0}^{M-1} ((2*k_2+1)*(floor(k_1/(2*M)^{k_2})\% (2*M)))} * \\
& (x+1)^{-\sum_{k_2=0}^{M-1} ((2*k_2+1)*(floor(k_1/(2*M)^{k_2})\% (2*M)))} * \\
& \Pi_{k_2=0}^{M-1} (factorial(floor(k_1/(2*M)^{k_2})\% (2*M))^{-1} * (2*k_2+1)^{-floor(k_1/(2*M)^{k_2})\% (2*M)}) \\
&)) \\
&)
\end{aligned}$$

kasutan multimomial astendajategenereerimiseks teist valemit $v[A] = \sum_{k_4=0}^{\infty} (k_2/2^{k_4*M+A}\%2)$. Ehk biti indexi jääk määrab, et mitmenda astendaja bit see on.

3.0.9 cos(ln(x)) maclaurin

cos:

$$\begin{aligned}
& \forall_x (x \in R \rightarrow \\
& cos(x) = \sum_{k=0}^{\infty} ((-1)^k * x^{2*k} * factorial(2*k)^{-1}) \\
&)
\end{aligned}$$

cos(y*ln(x)):

$$\begin{aligned}
& \forall_x (x > 0 \rightarrow \\
& sin(y*ln(x)) = \sum_{k_1=0}^{\infty} ((-1)^{k_1} * (2 * \sum_{k_2=0}^{\infty} ((2*k_2+1)^{-1} * (x-1)^{2*k_2+1} * (x+1)^{-2*k_2-1}))^{2*k_1} * y^{2*k_1} * \Gamma(2*k_1+1)^{-1}) \\
&)
\end{aligned}$$

Valem cos_y_ln_x_multinomial:

$$\begin{aligned}
& \forall_x (x > 0 \rightarrow cos(y*ln(x)) = \\
& lim_{M \rightarrow \infty} (\sum_{k_1=0}^{M-1} (y^{2*k_1} * 2^{2*k_1} * (-1)^{k_1} * \sum_{k_2=0}^{(2*k_1+1)^{M-1}} (\\
& (\sum_{k_3=0}^{M-1} (floor(k_2/(2*k_1+1)^{k_3})\% (2*k_1+1)) = 2*k_1) * \\
& (x-1)^{\sum_{k_3=0}^{M-1} ((2*k_3+1)*(floor(k_2/(2*k_1+1)^{k_3})\% (2*k_1+1)))} * \\
& (x+1)^{-\sum_{k_3=0}^{M-1} ((2*k_3+1)*(floor(k_2/(2*k_1+1)^{k_3})\% (2*k_1+1)))} * \\
& \Pi_{k_3=0}^{M-1} (factorial(floor(k_2/(2*k_1+1)^{k_3})\% (2*k_1+1))^{-1} * (2*k_3+1)^{-floor(k_2/(2*k_1+1)^{k_3})\% (2*k_1+1)}) \\
&))) \\
&)
\end{aligned}$$

Valem cos_y_ln_x_multinomial_2:

SIIN VIGA!

$$\begin{aligned}
& \forall_x (x > 0 \rightarrow sin(y*ln(x)) = \\
& lim_{M \rightarrow \infty} (\sum_{k_1=0}^{(2*M)^{M-1}} (\\
& (\sum_{k_2=0}^{M-1} ((floor(k_1/(2*M-1)^{k_2})\% (2*M-1)) + 1)\%2) * \\
& (\sum_{k_2=0}^{M-1} (floor(k_1/(2*M-1)^{k_2})\% (2*M-1)) < 2*M-1) * \\
& (y*2)^{\sum_{k_2=0}^{M-1} (floor(k_1/(2*M-1)^{k_2})\% (2*M-1))} * \\
& (-1)^{\sum_{k_2=0}^{M-1} (floor(k_1/(2*M-1)^{k_2})\% (2*M-1))/2} * \\
& (x-1)^{\sum_{k_2=0}^{M-1} ((2*k_2+1)*(floor(k_1/(2*M-1)^{k_2})\% (2*M-1)))} * \\
& (x+1)^{-\sum_{k_2=0}^{M-1} ((2*k_2+1)*(floor(k_1/(2*M-1)^{k_2})\% (2*M-1)))} * \\
& \Pi_{k_2=0}^{M-1} (factorial(floor(k_1/(2*M-1)^{k_2})\% (2*M-1))^{-1} * (2*k_2+1)^{-floor(k_1/(2*M-1)^{k_2})\% (2*M-1)}) \\
&))/2 \\
&)
\end{aligned}$$

veel üks valem:

SIIN VIGA!

$$\begin{aligned}
& \forall_x (x > 0 \rightarrow cos(y*ln(x)) = \\
& lim_{M \rightarrow \infty} (\sum_{n_1=0}^{(2*M)^{M-1}} (\\
& (\sum_{n_2=0}^{M-1} (floor(n_1/(2*M)^{n_2})\% (2*M))\%2) * \\
& (\sum_{k_2=0}^{M-1} (floor(k_1/(2*M-1)^{k_2})\% (2*M-1)) < 2*M-1) * \\
& (y*2)^{\sum_{n_2=0}^{M-1} (floor(n_1/(2*M)^{n_2})\% (2*M))} * \\
&)
\end{aligned}$$

$$\begin{aligned}
& (-1)^{\sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)) / 2} * \\
& (x-1)^{\sum_{n_2=0}^{M-1} ((2*n_2+1) * (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)))} * \\
& (x+1)^{-\sum_{n_2=0}^{M-1} ((2*n_2+1) * (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)))} * \\
& \Pi_{n_2=0}^{M-1} (\text{factorial}(\text{floor}(n_1/(2*M)^{n_2}) \% (2*M))^{-1} * (2*n_2+1)^{-\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)}) \\
&)) \\
&)
\end{aligned}$$

Valem cos_y_ln_x_multinomial_4_2:

SIIN VIGA!

$$\begin{aligned}
& \forall_x (x > 0 \rightarrow \cos(y * \ln(x)) = \\
& \lim_{M \rightarrow \infty} (\sum_{n_1=0}^{(2*M)^M-1} (\\
& ((1 + \\
& (y * 2)^{\sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M))} * \\
& (-1)^{\sum_{n_2=0}^{M-1} (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)) / 2} * \\
& (x-1)^{\sum_{n_2=0}^{M-1} ((2*n_2+1) * (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)))} * \\
& (x+1)^{-\sum_{n_2=0}^{M-1} ((2*n_2+1) * (\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)))} * \\
& \Pi_{n_2=0}^{M-1} (\text{factorial}(\text{floor}(n_1/(2*M)^{n_2}) \% (2*M))^{-1} * (2*n_2+1)^{-\text{floor}(n_1/(2*M)^{n_2}) \% (2*M)}) \\
&)) \\
&)
\end{aligned}$$

3.0.10 sisemine summa

$$A = \sum_{n_2=0}^{M-1} ((2*n_2+1) * (\text{floor}(n_0/(2*M)^{n_2}) \% (2*M)))$$

$$\begin{aligned}
& - \\
& \sum_{n_1=0}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})-1} (\\
& n_1^A * \\
& (n_1+1)^{-x_1} * \\
& (n_1+2)^{-A} * \\
&)
\end{aligned}$$

kasutan binomial valemite: $\forall_n (\forall_a (\forall_b ((a+b)^n = \sum_{k=0}^{\infty} (\text{faktorial}(n) * \text{faktorial}(k)^{-1} * \text{faktorial}(n-k)^{-1} * a^k * b^{n-k}))$

saadud tulemus:

$$\begin{aligned}
& \sum_{n_1=0}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})-1} (\\
& \sum_{k=0}^{\infty} (\text{faktorial}(-x_1) * \text{faktorial}(k)^{-1} * \text{faktorial}(-x_1-k)^{-1} * n_1^{k+A}) * \\
& \sum_{k=0}^{\infty} (\text{faktorial}(-A) * \text{faktorial}(k)^{-1} * \text{faktorial}(-A-k)^{-1} * n_1^k * 2^{-A-k}) * \\
&)
\end{aligned}$$

panen 1 summa 2 sisse:

$$\begin{aligned}
& \sum_{n_1=0}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})-1} (\\
& \sum_{k_1=0}^{\infty} (\sum_{k_2=0}^{\infty} (\text{faktorial}(-x_1) * \text{faktorial}(k_1)^{-1} * \text{faktorial}(-x_1-k_1)^{-1} * n_1^{k_1+A} * \text{faktorial}(-A) * \text{faktorial}(k_2)^{-1} * \\
& \text{faktorial}(-A-k_2)^{-1} * n_1^{k_2} * 2^{-A-k_2})) * \\
&)
\end{aligned}$$

lihtsustan seest ja viin n1 summa sisse:

$$\begin{aligned}
& \sum_{k_1=0}^{\infty} (\sum_{k_2=0}^{\infty} (\text{faktorial}(-x_1) * \text{faktorial}(k_1)^{-1} * \text{faktorial}(-x_1-k_1)^{-1} * \text{faktorial}(-A) * \text{faktorial}(k_2)^{-1} * \text{faktorial}(-A- \\
& k_2)^{-1} * 2^{-A-k_2} * \\
& \sum_{n_1=0}^{\text{floor}(e^{(\text{floor}(M)/\text{abs}(x_2)+q_2/x_2)*2\pi})-1} (n_1^{k_1+k_2+A}) \\
&))
\end{aligned}$$

3.0.11 veel

$$\int_a^b (dx * \sin(\ln(x))) = (-1/2 * b * (\cos(\log(b)) - \sin(\log(b)))) - (-1/2 * a * (\cos(\log(a)) - \sin(\log(a)))) = (b/2 * (\sin(\ln(b)) - \cos(\ln(b)))) - (a/2 * (\sin(\ln(a)) - \cos(\ln(a)))) = 2^{-1/2} * (b * \sin(\ln(b) - 2\pi/8) - a * \sin(\ln(a) - 2\pi/8)) = 2^{-1/2} * (a * \cos(\ln(a)) - b * \cos(\ln(b)))$$

$$\text{veel} \int_a^{a+b} (dx * \sin(\ln(x))) = 2^{-1/2} * (a * \cos(\ln(a)) - (a+b) * \cos(\ln(a+b))) = 2^{-1/2} * (a * (\cos(\ln(a)) - \cos(\ln(a+b))) - b * \cos(\ln(a+b)))$$

$$\text{veel} \int_{\text{floor}(e^{(\text{floor}(M)/2+1/4)*2\pi/\text{abs}(x_2)-\arctan((1-x_1)/x_2)/x_2})}^{\text{floor}(e^{(\text{floor}(M)/2+1/4)*2\pi/\text{abs}(x_2)-\arctan((1-x_1)/x_2)/x_2})+b} (dx * \sin(\ln(x))) = 2^{-1/2} * (\text{floor}(e^{(\text{floor}(M)/2+1/4)*2\pi/\text{abs}(x_2)-\arctan((1-x_1)/x_2)/x_2}) * (\cos(\ln(\text{floor}(e^{(\text{floor}(M)/2+1/4)*2\pi/\text{abs}(x_2)-\arctan((1-x_1)/x_2)/x_2}))) - \cos(\ln(\text{floor}(e^{(\text{floor}(M)/2+1/4)*2\pi/\text{abs}(x_2)-\arctan((1-x_1)/x_2)/x_2}) + b))) - b * \cos(\ln(\text{floor}(e^{(\text{floor}(M)/2+1/4)*2\pi/\text{abs}(x_2)-\arctan((1-x_1)/x_2)/x_2}) + b))) = 2^{-1/2} * (\text{floor}(e^{(\text{floor}(M)/2+1/4)*2\pi/\text{abs}(x_2)-\arctan((1-x_1)/x_2)/x_2}) * (\cos((\text{floor}(M)/2 + 1/4) * 2\pi/\text{abs}(x_2) - \arctan((1-x_1)/x_2)/x_2) - \cos(\ln(\text{floor}(e^{(\text{floor}(M)/2+1/4)*2\pi/\text{abs}(x_2)-\arctan((1-x_1)/x_2)/x_2}) + b))) - b * \cos(\ln(\text{floor}(e^{(\text{floor}(M)/2+1/4)*2\pi/\text{abs}(x_2)-\arctan((1-x_1)/x_2)/x_2}) + b))))$$

3.0.12 integral

$$\begin{aligned} \text{siinuse oma} \quad & \int_A^B (dy * \sin(x_2 * \ln(y) + (2\pi) * 2/4) * y^{-x_1}) = \\ & - \int_A^B (dy * \sin(x_2 * \ln(y)) * y^{-x_1}) = \\ & \frac{B^{1-x_1} * ((x_1-1) * \sin(x_2 * \ln(B)) + x_2 * \cos(x_2 * \ln(B)))}{x_1^2 - 2 * x_1 + x_2^2 + 1} - \frac{A^{1-x_1} * ((x_1-1) * \sin(x_2 * \ln(A)) + x_2 * \cos(x_2 * \ln(A)))}{x_1^2 - 2 * x_1 + x_2^2 + 1} = \\ & (B^{1-x_1} * ((x_1-1) * \sin(x_2 * \ln(B)) + x_2 * \cos(x_2 * \ln(B))) - A^{1-x_1} * ((x_1-1) * \sin(x_2 * \ln(A)) + x_2 * \cos(x_2 * \ln(A)))) * \\ & ((1-x_1)^2 + x_2^2)^{-1} = \\ & (B^{1-x_1} * (x_1-1) * \sin(x_2 * \ln(B)) + B^{1-x_1} * x_2 * \cos(x_2 * \ln(B)) - A^{1-x_1} * (x_1-1) * \sin(x_2 * \ln(A)) - A^{1-x_1} * x_2 * \cos(x_2 * \ln(A))) * \\ & ((1-x_1)^2 + x_2^2)^{-1} = \\ & (-B^{1-x_1} * (1-x_1) * \sin(x_2 * \ln(B)) + B^{1-x_1} * x_2 * \cos(x_2 * \ln(B)) + A^{1-x_1} * (1-x_1) * \sin(x_2 * \ln(A)) - A^{1-x_1} * x_2 * \cos(x_2 * \ln(A))) * \\ & ((1-x_1)^2 + x_2^2)^{-1} = \\ & (B^{1-x_1} * (-(1-x_1) * \sin(x_2 * \ln(B)) + x_2 * \cos(x_2 * \ln(B))) + A^{1-x_1} * ((1-x_1) * \sin(x_2 * \ln(A)) - x_2 * \cos(x_2 * \ln(A)))) * \\ & ((1-x_1)^2 + x_2^2)^{-1} = \\ & (B^{1-x_1} * \sqrt{((1-x_1)^2 + x_2^2)} * \sin(x_2 * \ln(B) - \arctan(x_2/(1-x_1))) + A^{1-x_1} * \sqrt{((1-x_1)^2 + x_2^2)} * \sin(x_2 * \ln(A) - \arctan(x_2/(1-x_1)))) * \\ & ((1-x_1)^2 + x_2^2)^{-1} = \\ & (B^{1-x_1} * \sin(x_2 * \ln(B) - \arctan(x_2/(1-x_1))) + A^{1-x_1} * \sin(x_2 * \ln(A) - \arctan(x_2/(1-x_1)))) * ((1-x_1)^2 + x_2^2)^{-1/2} \end{aligned}$$

$$\begin{aligned} \text{coosinuse oma} \quad & \int_A^B (dy * \sin(x_2 * \ln(y) + (2\pi) * 1/4) * y^{-x_1}) = \\ & \int_A^B (dy * \cos(x_2 * \ln(y)) * y^{-x_1}) = \\ & \frac{B^{1-x_1} * (x_2 * \sin(x_2 * \ln(B)) - (x_1-1) * \cos(x_2 * \ln(B)))}{x_1^2 - 2 * x_1 + x_2^2 + 1} - \frac{A^{1-x_1} * (x_2 * \sin(x_2 * \ln(A)) - (x_1-1) * \cos(x_2 * \ln(A)))}{x_1^2 - 2 * x_1 + x_2^2 + 1} = \\ & (B^{1-x_1} * (x_2 * \sin(x_2 * \ln(B)) - (x_1-1) * \cos(x_2 * \ln(B))) - A^{1-x_1} * (x_2 * \sin(x_2 * \ln(A)) - (x_1-1) * \cos(x_2 * \ln(A)))) * \\ & ((1-x_1)^2 + x_2^2)^{-1} = \\ & (B^{1-x_1} * x_2 * \sin(x_2 * \ln(B)) - B^{1-x_1} * (x_1-1) * \cos(x_2 * \ln(B)) - A^{1-x_1} * x_2 * \sin(x_2 * \ln(A)) + A^{1-x_1} * (x_1-1) * \cos(x_2 * \ln(A))) * \\ & ((1-x_1)^2 + x_2^2)^{-1} = \\ & (B^{1-x_1} * x_2 * \sin(x_2 * \ln(B)) + B^{1-x_1} * (1-x_1) * \cos(x_2 * \ln(B)) - A^{1-x_1} * x_2 * \sin(x_2 * \ln(A)) - A^{1-x_1} * (1-x_1) * \cos(x_2 * \ln(A))) * \\ & ((1-x_1)^2 + x_2^2)^{-1} = \\ & (B^{1-x_1} * (x_2 * \sin(x_2 * \ln(B)) + (1-x_1) * \cos(x_2 * \ln(B))) - A^{1-x_1} * (x_2 * \sin(x_2 * \ln(A)) - (1-x_1) * \cos(x_2 * \ln(A)))) * \\ & ((1-x_1)^2 + x_2^2)^{-1} = \\ & (B^{1-x_1} * \sin(x_2 * \ln(B) + \arctan((1-x_1)/x_2)) - A^{1-x_1} * \sin(x_2 * \ln(A) + \arctan((1-x_1)/x_2))) * ((1-x_1)^2 + x_2^2)^{-1/2} \end{aligned}$$